Self Playing Guitar Amplifier Lab Report

ELT-213-20456
Active Circuits Design

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Electrical & Computer Engineering

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Part 1 Intro:

Our group is working on a guitar amp project to assist musicians. We're building an amp that can accompany the guitarist with a sustained drone note, giving them a whole new level of creativity to play with. From what we've researched, there isn't a commercial guitar amp out there that offers this feature, so this idea is completely original.

We're using two LM386 amplifiers and an ICL8038 tunable wave generator to make this happen. The wave generator will be set to frequencies between 200-2000Hz, which gives us just over three octaves of range. One LM386 amp will be dedicated to powering the guitar, while the other will amplify the function generator's signal, creating the drone note sound.

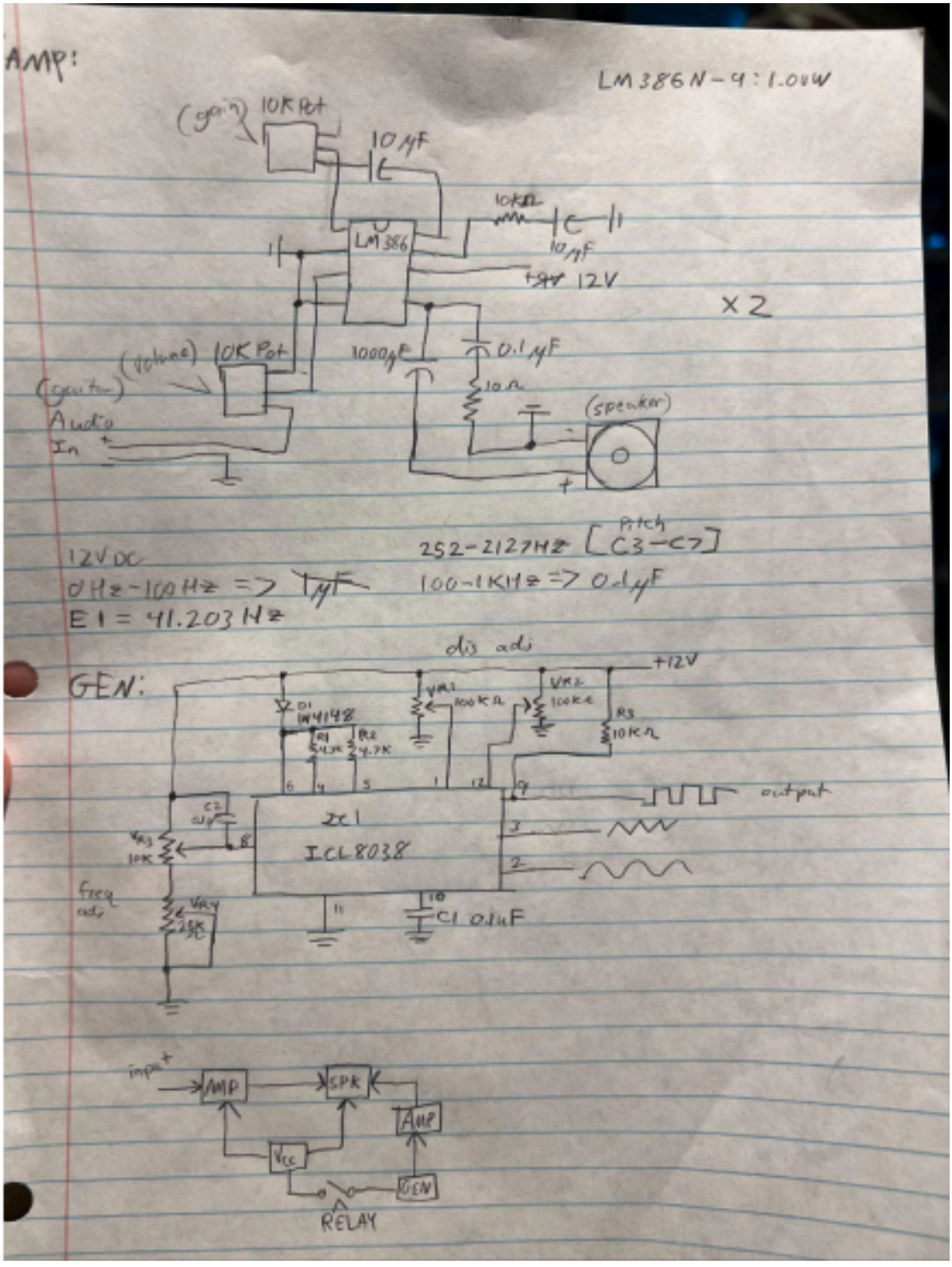
The ICL8038 frequency generator is a key component here, allowing us to dial in the perfect frequency for the drone note. With tone and timbre control, the guitarist will have a constant, harmonically-rich sound to play over.

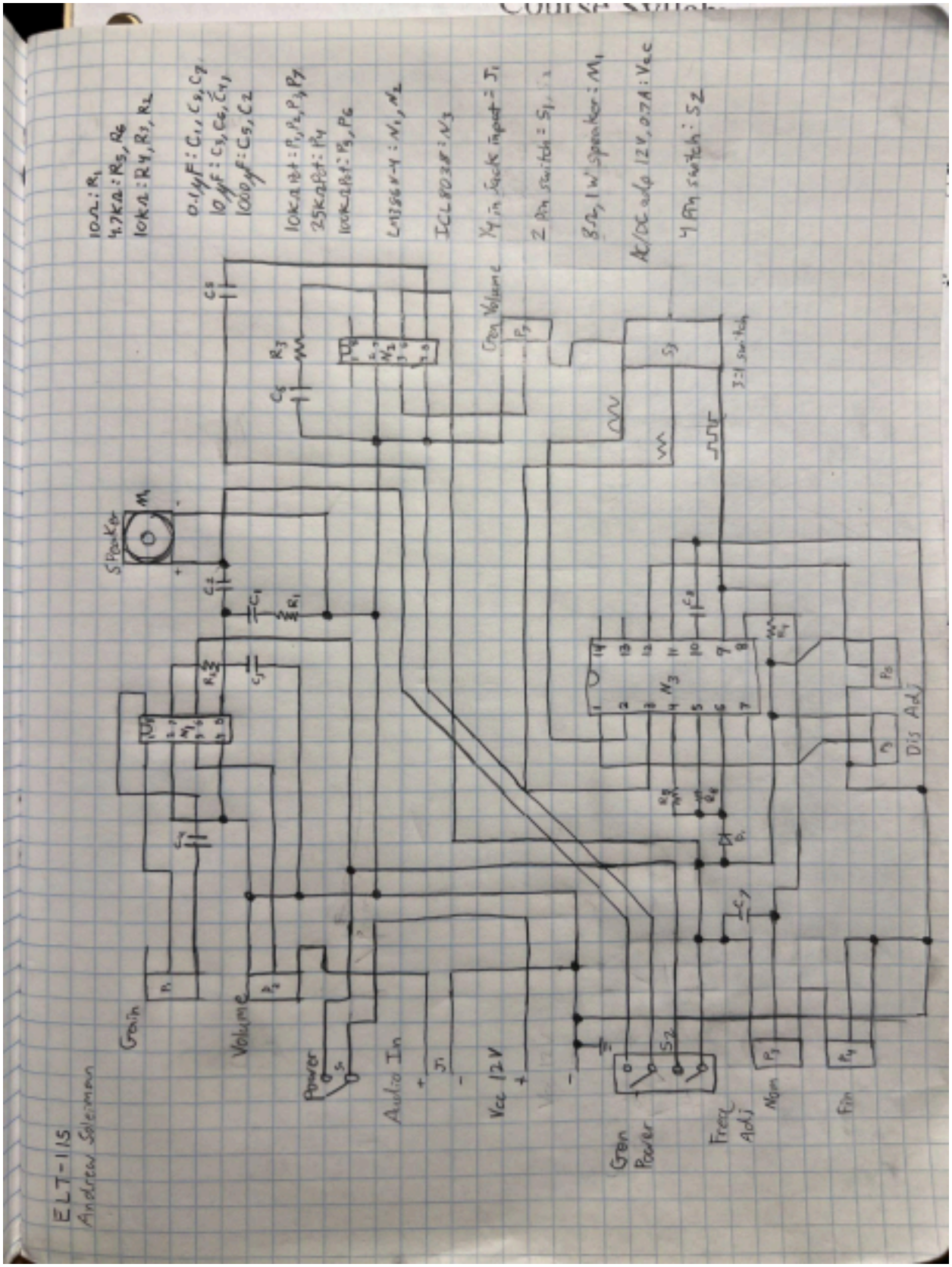
Parts List:

- LM386N-4-ND x 2 (Amplifier IC 1-Channel (Mono) Class AB 8-PDIP)
- ICL8038CCPD-ND x 1 (Precision Waveform Generator/Voltage Controlled Oscillator) ● 987-1710-ND x 4 (POT 10K OHM 1/20W PLASTIC LINEAR)
- 987-1303-ND x 1 (POT 25K OHM 1/5W PLASTIC LINEAR)
- PDB181-K420K-104A2-ND x 2 (POT 100K OHM 1/10W CARBON LOG)
- 668-1236-ND x 2 (a 3” 8ohm, 1watt speaker would be ideal, however, it’s not on digikey)
- 4506-PWWR0013F10R0K9-ND x 1(10 Ohms ±1% 13.5W Through Hole Resistor Axial Wirewound)
- OY472KE-ND x 2 (4.7k ohm resistor)
- CF14JT10K0TR-ND x 3 (10k ohm resistor)
- 104CKR050M-ND x 3 (0.1uF capacitor)
- 1189-1894-ND x 3 (10uF capacitor)
- 1189-1873-ND x 2 (1000uF capacitor)
- 237-1453-ND x 1 (AC/DC WALL MOUNT ADAPTER 12V, 0.7A, 8.4W)

- SC2014-ND x 1 (6.35mm (0.250", 1/4") - Headphone Phone Jack Mono Connector Solder)
- 2449-ANT21SECQE-ND x 1 (Toggle Switch DPDT Through Hole)
- 2449-ANT11SECQE-ND x 1 (Toggle Switch SPDT Panel Mount)
- CKN9822-ND x 1 (Rotary Switch 3 Position SP3T 2.5A (AC), 350mA (DC) 125 VAC Panel Mount)

Schematics/ Designs:




ELT-115
Andrew Soleiman
Gain
Volume
Power
Audio In
Vcc 12V
Gen Power
Freq Adj
Fine
Dis Adj
Course Switch
Gen Volume
Speaker
3:1 switch
100Ω: R1
4.7KΩ: R5, R6
10KΩ: R4, R3, R2
0.1μF: C1, C3, C8
10μF: C5, C6, C4
1000μF: C5, C2
10KΩ pot: P1, P2, P3, P7
25KΩ pot: P4
100KΩ pot: P5, P6
LM386-N: N1, N2
ICL8038: N3
4 pin socket: J1
2 Pin switch: S1
8Ω, 1W Speaker: M1
AC/DC adp. 12V, 0.7A: Vcc
4 Pin switch: S2

Picture 1 shows both the basic amplifiers and function generator schematics, and picture 2 shows them incorporated together.

N1 (LM386 amp number 1) is used purely for the guitar input/ output. Audio In goes through the main volume potentiometer and into pin 3 of N1. Pin 7 of N1 goes into the output speaker. With some added capacitors and low load resistors to filter out high and low frequencies, N1 works as the basic guitar amp.

N3 (ICL8083 Frequency Generator) is how the drone note will be achieved. By using a 0.1uF capacitor from pin 18 of N3 to ground, the frequency range will be approximately 200-2000Hz, giving us a melodic range of G3(196Hz) to B6(1975.5Hz), or a little over three octaves. N3 also has three types of outputs: square wave, triangular wave, and sine wave. Using a 3-1 switch, we can choose the desired output wave, each one giving a different timbre (different tone or sound). This output wave is then passed through a volume control potentiometer and into pin 3 of N2.

N2 (LM386 amp number 2) is used to amplify the generator's output wave and feed it into the output speaker.

Assembly:

Assembling the LM386 amplifiers:

- Gather components for both amps (as shown in schematic):
    - LM386 IC
    - 12V power supply
    - Resistors: 10kΩ (2), 100Ω, 10Ω
    - Capacitors: 0.1µF (3), 10µF (2), 100µF, 470pF, 1000µF
    - Potentiometers: 10kΩ Gain, 10kΩ Volume
    - Speaker (8Ω)
    - Breadboard & wires
- Prepare breadboard:
    - Insert LM386 IC in center of breadboard (ensure notch faces correct orientation).
- Power connections:
    - Connect +9V battery to positive rail; ground to negative rail.
    - Place 100µF capacitor between +Vcc (pin 6) and GND (pin 4) for decoupling.
    - Add 0.1µF capacitor near pin 6 for high-frequency bypass.

- Input stage:
    - Connect “Audio In” to 10kΩ Volume pot (one end to input, other to GND, middle to LM386 pin 3 via 0.1μF capacitor).
    - Connect 470pF capacitor between pin 3 and GND (optional stability).
- Gain setup:
    - Connect 10kΩ Gain pot between pin 1 and pin 8 (adjusts gain).
    - Place 10μF capacitor between pin 1 and GND.
- Output stage:
    - Connect pin 5 (output) to 1000μF capacitor (positive side to pin 5).
    - Connect the other side of 1000μF capacitor to + speaker terminal.
    - Connect speaker – terminal to GND.
    - Add 10Ω resistor + 0.1μF capacitor in series across speaker
- Repeat
    - Repeat steps to build two amps.

Assembling the ICL8083 function generator:

- Gather components (refer to the schematic):
    - IC1: ICL8038 (N3).
    - Resistors: R1 = 4.7 kΩ, R2 = 4.7 kΩ, R3 = 10 kΩ.
    - Variable resistors: VR1 = 100 kΩ, VR2 = 100 kΩ, VR3 = 10 kΩ, VR4 = 25 kΩ.
    - Capacitors: C1 = 0.1 μF, C2 = 0.1 μF.
    - Diode: 1N4148 (D1).
    - Power: 12 V supply.
- Prepare the PCB or breadboard:
    - Lay out the components matching the schematic connections.
    - Use a clean ground (GND) plane for all “-” lines.
- Install the IC:
    - Place the ICL8038 in the socket (pin 1 marked).
    - Orient the notch or dot toward the top of the schematic (pin 1 → top-left).
- Wire the power & diode:
    - Connect +12 V to the top rail and GND to the bottom rail.
    - Insert D1 between the 12 V line and VR1/VR2, anode to VR1, cathode to 12 V.

- Add resistors & variable resistors:
    - Connect R1 & R2 between pins 4/5 and the IC's internal bias network.
    - Wire VR1 & VR2 (100 kΩ) for distortion adjustment, linking them to R3 (10 kΩ).
    - Hook VR3 (10 kΩ) to pin 8 for "Normal" frequency setting.
    - Hook VR4 (25 kΩ) for "Fine" frequency adjustment, linking it to VR3.
- Install capacitors:
    - Place C1 = 0.1 µF between pin 18 and GND (replacing the original 0.01 µF).
    - Place C2 = 0.1 µF between pin 8 and GND.

Complete the Guitar Amp:

- Gather components:
    - N1, N2, N3
    - Power Switches
    - Audio Jack
    - Power Supply
- Prepare PCB:
    - Place all parts on PCB according to the schematic
    - Solder all connections and begin testing

Testing Steps:

Using a multimeter and an oscilloscope, the system can be tested to ensure it's working properly. Each component will be tested separately.

Testing N1 and N2 amps:

- Gather tools: A multimeter and a guitar or audio source
- Inspect the circuit: double-check the schematic and verify all components are placed correctly on the breadboard. Make sure the LM386 pinout matches the layout (pin 1 = gain, pin 2 = –input, pin 3 = +input, pin 5 = output, pin 6 = V S).
- Power up: connect the 12V power supply to the positive and negative rails as shown. Measure the voltage at the LM386 VS pin to confirm you have ~12V.

- Check the bias: with no input signal, measure the DC voltage at the output (pin 5). It should be roughly half the supply voltage (≈6V) for proper bias.
- Test the audio path: plug your guitar or a line-out audio source into the “Audio In” jack. Set the gain and volume pots to mid-range.
- Listen: connect a 4Ω speaker to the output (pin 5 via the 1000μF capacitor). Play a note on the guitar. You should hear amplified sound. Adjust the 10kΩ gain pot to change the amplification level and the 10kΩ volume pot to control the output level.
- Troubleshoot: if there’s no sound, check for:
    - loose or wrong connections
    - faulty LM386 or shorted pins
    - capacitor polarity (electrolytic caps must be oriented correctly)
    - speaker wiring
- Final check: use the multimeter to measure AC voltage at the speaker terminals while playing a signal; you should see a varying voltage indicating the amp is working.

Testing N3 frequency generator:

- Gather tools: you need a 12 V DC power supply, a multimeter, an oscilloscope (or a frequency counter), and a breadboard with jumper wires.
- Check the circuit: Check the orientation of the diode D1 and capacitors C1, C2.
- Power up: connect the 12 V DC supply to the +12 V rail and ground. Measure the supply voltage at the IC’s V+ pin to confirm 12 V.
- Set frequency: adjust the “Frequency Adjust” potentiometers VR3 (10 kΩ) and VR4 (25 kΩ) to mid-range for an initial ~1 kHz output. The fine pot VR4 lets you tweak the frequency precisely.
- Check waveforms: attach the oscilloscope probes to the three output pins (2, 3, 9). You should see:
    - Pin 2: sine wave (Sine Output).
    - Pin 3: triangle wave (Triangular Output).
    - Pin 9: square wave (Square Output).
- Measure frequency: use the oscilloscope or frequency counter on any output to verify the frequency is near your desired value.
- Test distortion: turn the “Distortion” potentiometers VR1 and VR2 to vary the waveform symmetry. Observe changes in the sine wave purity on the oscilloscope.

- Verify amplitude: check that the output voltage swing matches the expected levels for the 12 V supply (typically ~6 Vpp for sine/triangle, full swing for square).
- Troubleshoot: if no output or distorted signals appear, inspect wiring, component values, or replace the ICL8038 if faulty.

Troubleshooting/ Possible Issues:

Since this idea is completely original, there will most likely be issues/ design flaws that will need to be addressed/ solved. Some of the predicted problems include:

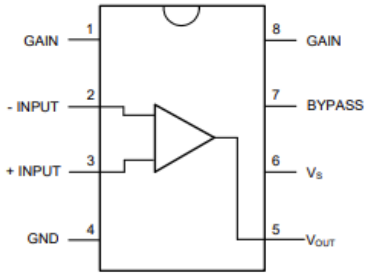
- Possible Input Issues:
    - The schematics have two audio outputs leading into one speaker input. This could potentially cause a feedback or noise issue.
- Underpowered Signals:
    - The output signal of the frequency generator has an unclear peak voltage. This could lead to underpowered sine waves resulting in a low volume for the drone note.
- Underpowered Power Supply
    - Although both the LM386 and ICL8083 chip can function on only 9 volts, powering two LM386 amplifier chips and a ICL8083 frequency generator chip with one 12 volt power supply could lead to some underpowered circuits.
- Underpowered Speaker
    - The speaker used in this system is relatively small and low quality. This speaker also has a frequency range of 450-20000Hz, meaning the low end waves will not be heard. This could lead to poor noise quality.
- Unclear Frequency Range
    - After multiple calculations, the predicted frequency range of the ICL8083 generator is 200-2000Hz. However, those calculations did not include the specific power supply or the LM386 amplifiers. This could result in a different frequency range than what was calculated.

Data:

LM386 Data Sheet:

5 Pin Configuration and Functions

D Package
8-Pin MSOP
Top View



Pin Functions

PIN		TYPE	DESCRIPTION
NAME	NO.		
GAIN	1	–	Gain setting pin
–INPUT	2	I	Inverting input
+INPUT	3	I	Noninverting input
GND	4	P	Ground reference
V_{OUT}	5	O	Output
V_S	6	P	Power supply voltage
BYPASS	7	O	Bypass decoupling path
GAIN	8	–	Gain setting pin

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

		MIN	MAX	UNIT
Supply Voltage, V_{CC}	LM386N-1/-3, LM386M-1		15	V
	LM386N-4		22	
Package Dissipation	LM386N		1.25	W
	LM386M		0.73	
	LM386MM-1		0.595	
Input Voltage, V_I		–0.4	0.4	V
Storage temperature, T_{stg}		–65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

6.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)	±1000	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101(2)	±1000	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

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ICL8083 Data Sheet:

RENESAS

DATASHEET

OBSOLETE PRODUCT
NO RECOMMENDED REPLACEMENT
contact our Technical Support Center at
1-888-INTERSIL or www.intersil.com/tsc

ICL8038

Precision Waveform Generator/Voltage Controlled Oscillator

FN2864
Rev.4.00
Apr 2001

The ICL8038 waveform generator is a monolithic integrated circuit capable of producing high accuracy sine, square, triangular, sawtooth and pulse waveforms with a minimum of external components. The frequency (or repetition rate) can be selected externally from 0.001Hz to more than 300kHz using either resistors or capacitors, and frequency modulation and sweeping can be accomplished with an external voltage. The ICL8038 is fabricated with advanced monolithic technology, using Schottky barrier diodes and thin film resistors, and the output is stable over a wide range of temperature and supply variations. These devices may be interfaced with phase locked loop circuitry to reduce temperature drift to less than 250ppm/°C.

Features

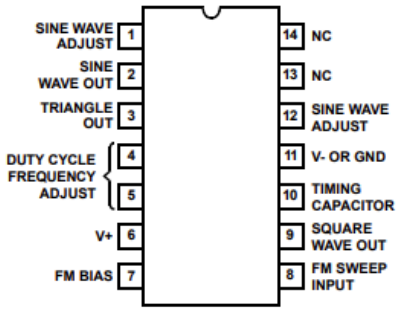
- Low Frequency Drift with Temperature. . . . . 250ppm/°C
- Low Distortion . . . . . 1% (Sine Wave Output)
- High Linearity . . . . . .0.1% (Triangle Wave Output)
- Wide Frequency Range . . . . . .0.001Hz to 300kHz
- Variable Duty Cycle . . . . . 2% to 98%
- High Level Outputs. . . . . TTL to 28V
- Simultaneous Sine, Square, and Triangle Wave Outputs
- Easy to Use - Just a Handful of External Components Required

Ordering Information

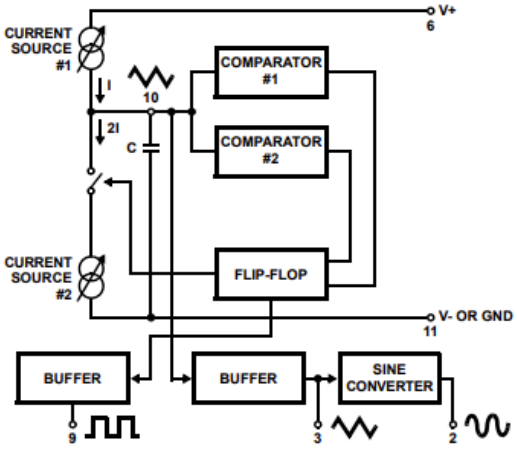
PART NUMBER	STABILITY	TEMP. RANGE (°C)	PACKAGE	PKG. NO.
ICL8038CCPD	250ppm/°C (Typ)	0 to 70	14 Ld PDIP	E14.3
ICL8038CCJD	250ppm/°C (Typ)	0 to 70	14 Ld CERDIP	F14.3
ICL8038BCJD	180ppm/°C (Typ)	0 to 70	14 Ld CERDIP	F14.3
ICL8038ACJD	120ppm/°C (Typ)	0 to 70	14 Ld CERDIP	F14.3

Pinout

ICL8038
(PDIP, CERDIP)
TOP VIEW



Functional Diagram



668-1236-ND Speaker Data Sheet:

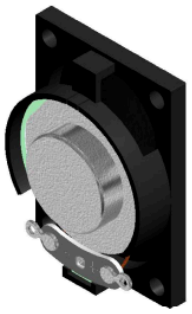
SPECIFICATIONS

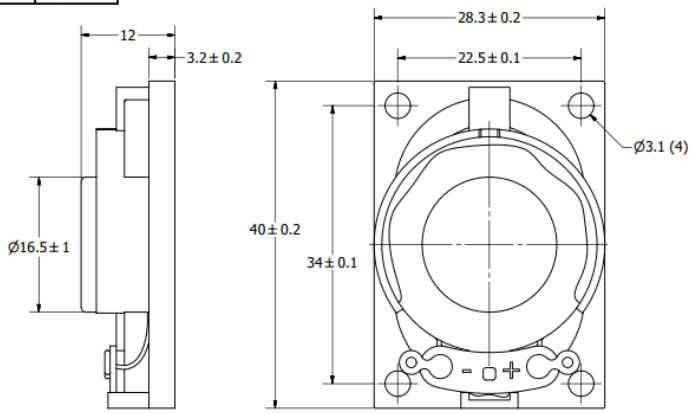
PARAMETERS	VALUES	UNITS
RATED INPUT POWER	1	W
MAX INPUT POWER	2	W
IMPEDANCE	8 ± 15%	OHM
OUTPUT SPL @ 1W/0.5M (AT 1.0, 1.2, 1.5, 2.0 kHz)	84 ± 3	dBA
DISTORTION (MAX.)	5%	-
RESONANT FREQUENCY	450 ± 20%	Hz
FREQUENCY RANGE	450 ~ 20000	Hz
HOUSING MATERIAL	ABS	-
CONE MATERIAL	PAPER	-
MAGNET MATERIAL	Nd-Fe-B	-
OPERATING TEMPERATURE	-20 ~ +60	°C
WEIGHT	13	grams

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REVISION HISTORY

LTR	DESCRIPTION	DATE	APPROVED
-	RELEASED FROM ENGINEERING	6/29/2006	
A	REVISED CONE MATERIAL	3/16/2007	R.W.
B	REVISED TO INVENTOR 3-D DRAWING TEMPLATE	5/19/2008	B.R.
C	REVISED SPEAKER HEIGHT	5/5/2015	M.L.





NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. SPECIFICATIONS SUBJECT TO CHANGE OR WITHDRAWL WITHOUT NOTICE.
3. THIS PART IS RoHS 2011/65/EU COMPLIANT.

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS, TOLERANCES ARE ±0.5 AND ANGLES ARE ±3°.	SIZE	Designed by	Date	Checked by	Date	Approved by	Date	Drawn Date
	A3	J.D.	6/28/2006	E.P.	6/28/2006	B.R.	6/28/2006	5/19/2008

pui audio inc — a projects unlimited company

AS04008PO-2-R

Speaker

AS04008PO-2-R.idw

Edition: -

Sheet: 1 / 1

WSU120-0700 Power Supply Data Sheet:

TRIAD MAGNETICS

Wall Plug - Ins

WSU120-0700



Electrical Specifications (@25C)

1. Input Voltage rating: 100-240VAC, 50-60Hz
2. Input Voltage range: 90-264VAC
3. Input current: <0.35A(RMS) @ 115VAC
4. Max Inrush Current: <60A peak @ 115VAC (Cold start)
5. Output Voltage: 12VDC
6. Output Current: .7A
7. Regulation (line & load): ±5%
8. Ripple & Noise: 150mVpk-pk Max
9. No load power (stand by): <100mW
10. Average Efficiency: ≥80.93%. Meets minimum level VI efficiency.

Environmental Specifications

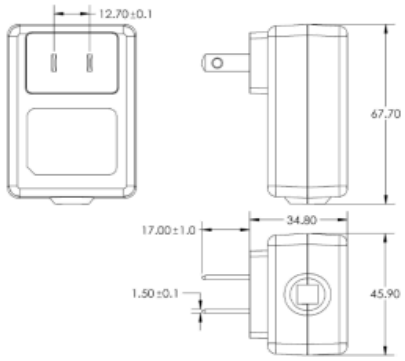
1. Operating Temperature Range: 0°C to +40° C @ full load
2. Storage Temperature Range: -20°C to 60°C
3. Humidity: 5% to 95%, Non-condensing

Reliability Specifications

1. Leakage Current: <0.25mA (264VAC)
2. Dielectric Strength (Hi-pot): 4242VDC/3secs., 5mA Max
3. Warranty: 5 years

Mechanical Parameters

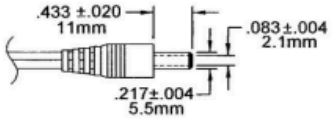
Case Type: Thermoplastic molded enclosure.
Output Cord: 24 AWG, 6 Ft. Long Nom.



Safety & EMI

UL: 62368-1 Class II, Double Insulated
UL File Number: E345519
EMI standard: FCC part15 Subpart B, ICES-003 Issue 6, ANSI C63.4-2014
Over voltage and short circuit protected

c UL us LISTED



EISA 2007/CEC Compliance: All WSU Power Supplies manufactured after February 10, 2016 will meet the minimum efficiency levels for direct operation as defined by DOE Docket Number EERE-2008-BT-STD-0005-0219. Triad's level VI products will have date code no later than 1605 (YYWW) where 16 is the year and 05 is the 5th week of 2016. In accordance with DOE requirement the label will also contain the Roman numeral VI with a circle.

RoHS Compliance: As of manufacturing date February 2016, all standard products meet the requirements of 2015/863/EU, known as the RoHS 3 initiative.

Upon printing, this document is considered "uncontrolled". Please contact Triad Magnetics' website for the most current version.



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Perris, California 92571

Publish Date: August 20, 2021

Project Schedule:

Depending on what next semester looks like/ what the curriculum is, our project schedule will be as follows:

1-2 Weeks	Wait for parts to arrive
1-2 Weeks	Assemble project
7-8 Weeks	Troubleshoot project
1-2 Weeks	Document project
2-3 Weeks	Final presentation of project

Conclusion:

In summary, this project proposes a self-playing guitar amp system that combines two LM386 amplifiers and an ICL8038 function generator to generate a sustained drone note, giving guitarists a new creative outlet to experiment with. The system's design aims to overcome traditional guitar amp limitations, offering a unique tool for sound enhancement. Further development and testing are needed to bring this concept to life and assess its performance.

Stage 2 Intro:

With the main objective stated and how it will be implemented and achieved, the second step of the fabrication process can start. This stage shifts from planning and part selection, to hands-on assembly and initial validation of each component.

The primary purpose of this stage is to:

- Build and test each main component of the project separately on the bench
    - LM386 amplifier circuit for basic signal gain and output drive
    - ICL8038 oscillator generator for the reference tone
    - Volume control using panel-mount potentiometers for each input channel
    - Main power switches for both circuits to allow independent testing
- Combine each part together into a single temporary test setup
- Fine tune the circuit values for stable output, clean mixing, and minimal noise
- Troubleshoot any problems that may occur during integration or power-up
- Order new parts if needed to replace delayed shipments or faulty components
- Measure different outputs of each circuit with a multimeter and oscilloscope for verification
- Document the process of stage 2 with photos, measurements, and build notes
- Finalize the schematics for stage 3 based on the tested and working values

With everything listed above completed, a final schematic can be drawn and the circuit can be assembled.

Troubleshooting the Individual Modules:

The LM386 amplifier was initially built using a breadboard, resistors, capacitors, and wires. All connections were placed according to the datasheet example circuit and checked for continuity before power was applied. A bench power supply was set to 12 volts and was used as the power source for the amplifier rails. A function generator was set to 455.6 Hz and 300 mV amplitude for the amplifier input. This frequency and voltage peak mimics an electric guitar input, specifically the note A4 at medium to loud volume.

The ICL8038 frequency generator chip was delayed in shipping and was not available for the initial build. An XR2206 frequency generator was ordered and built instead as a direct substitute. This XR2206 oscillator chip arrived as a complete kit and came with a custom PCB and all required circuit components to complete the generator. Each part was soldered together according to the provided schematics, and the completed board was tested for output. The unit was tuned to operate from 100 to 3,000 Hz using the onboard jumpers.

Both the LM386 amplifier and the XR2206 oscillator circuit were combined on the bench, with the frequency generator serving as the primary input and the LM386 as the signal amplifier. The function generator input, set to 455.6 Hz and 300 mV amplitude, was then added to the amplifier input as well through a simple summing node, resulting in a dual input configuration. An 8 ohm, 6 watt speaker was connected to the amplifier output and used as the load for testing.

The speaker's impedance was a good match to the circuit, but the wattage rating was larger than what the LM386 amplifier could supply. This does not cause an issue, as overshooting the wattage rating of the speaker does not result in distortion or added noise and simply leaves headroom. As long as the impedance is matched correctly, the speaker will work as normal and reproduce the combined signal without strain.

Problems that Developed:

The initial 12 volt supply was too high for the LM386 chip, causing major signal distortion at the output. The supply was lowered to 9 volts from a bench power supply, and later a +9 volt battery was used for testing instead. This change brought the operating voltage within the chip's recommended range and fixed the distortion issues.

The XR2206 output amplitude was much higher than expected from the kit documentation. A 500 mV to 1 V peak-to-peak output was expected at the default setting. However, using an oscilloscope to verify the waveform, the output was measured to be close to 8 volts peak-to-peak under no load.

While having both the XR2206 oscillator output and the function generator output connected together and fed into the amplifier input, the stronger signal would trump the lower amplitude signal, resulting in only one note being played at a time. The two sources were tied directly without isolation, so the frequencies would "fight" each other and partially cancel each other out at the summing point.

Two separate power sources were used while testing the combined circuit, a +12 volt power supply for the oscillator and a +9 volt battery for the amplifier. While building, all the grounds were tied together on the breadboard, resulting in ground loops and unintentional coupling between the power sources. This caused major distortion in both the LM386 amplifier circuit and the XR2206 oscillator circuit, and added noticeable hum to the speaker output.

Solutions for Problems:

Since the amplitude of the XR2206 oscillator was much higher than expected, the solution was simply to remove the secondary amplifier stage from the signal path. This reduced the overall gain on the oscillator side and made both the XR2206 generator output and the amplified guitar output match more closely in peak-to-peak values at the input.

To allow two or more frequencies to merge into one input without interference, a passive summing box was added before the LM386. This simple circuit does not require a power supply, as it only acts as a resistive network for multiple signals to combine without direct coupling or wave cancellation. Two 10k linear potentiometers and two 10k resistors were initially used to set the input levels and provide isolation between the sources.

To avoid using multiple power supplies and the consequential ground loop noise, a single +12 volt supply was used for the entire system instead. However, the LM386 amplifier performed best at +9 volts for clean output. A +12 volt to +9 volt linear regulator converter was added to drop the voltage cleanly for the amplifier stage alone.

The schematics were updated to match the new circuit layout, including the passive summing mixer for the guitar and oscillator inputs, a single LM386 amplifier for the mixed signal, and the added regulator for the supply rail. Fine tuned potentiometers were included for volume, frequency, and gain control to allow adjustment during testing and demonstration.

Implementation of the Solutions:

For better and cleaner audio, a better, higher quality speaker was bought and used. This helped reduce random hums, unwanted noise, and scratchy sounding audio.

Instead of using a step down buck converter for the +12 volt to +9 volt conversion, two separate power sources (a DC power supply for the +12 volts, and a DC battery for the +9 volts) were used with careful notice to not cross the different grounds. By keeping the DC power supply only for the XR2206 frequency generator, there were no ground loops, feedback, or odd noise or hums.

To build and tune a passive summing box, multiple resistors values were used for testing. Potentiometer limiting and tapering were needed as well to fine tune the mixing control. A 470k ohm resistor with a 10k logarithmic potentiometer for the oscillator output was ideal for matching the guitar volume with the XR2206 frequency output volume.

Solderboards were used to construct a fully integrated LM386 amplifier build. The first iteration worked well and passed initial testing, however it was large and nonoptimal as it used the majority of the board. A second board was built to be more condensed and to save space. However, the board was wired incorrectly and the amplifier did not function properly on power-up. A third and final amplifier was built with more precision and careful checking of each joint. This amp functioned as intended and was later used for the final build.

Building the LM386 Amp Board:

The assembly instructions listed in stage 1 were followed to build and solder the LM386 amps.

- Gather components for both amps (as shown in schematic):
    - LM386 IC
    - 12V power supply
    - Resistors: 10kΩ (2), 100Ω, 10Ω
    - Capacitors: 0.1µF (3), 10µF (2), 100µF, 470pF, 1000µF
    - Potentiometers: 10kΩ Gain, 10kΩ Volume
    - Speaker (8Ω)
    - Solderboard & wires
- Prepare breadboard:
    - Insert LM386 IC in center of solderboard (ensure notch faces correct orientation).
- Power connections:
    - Connect +9V battery to positive rail; ground to negative rail.
    - Place 100µF capacitor between +Vcc (pin 6) and GND (pin 4) for decoupling.
    - Add 0.1µF capacitor near pin 6 for high-frequency bypass.
- Input stage:
    - Connect “Audio In” to a 10kΩ Log Volume pot (one end to input, other to GND, middle to LM386 pin 3 via 0.1µF capacitor).
    - Connect 470pF capacitor between pin 3 and GND (optional stability).
- Gain setup:
    - Connect 10kΩ Gain linear pot between pin 1 and pin 8 (adjusts gain).
    - Place 10µF capacitor between pin 1 and GND.
- Output stage:
    - Connect pin 5 (output) to 1000µF capacitor (positive side to pin 5).
    - 22Connect the other side of 1000µF capacitor to + speaker terminal.
    - Connect speaker – terminal to GND.
    - Add 10Ω resistor + 0.1µF capacitor in series across speaker

Building the XR2206 Function Generator:

The assembly instructions listed in stage 1 were followed to build and solder the XR2206 generator.

- Gather components for kit (as shown on PCB):
    - XR2206 IC and 16-pin socket
    - Resistors: 1kΩ, 5.1kΩ (3), 330Ω
    - Capacitors: 100µF, 10µF (2), 0.1µF, 1µF, 47nF, 2.2nF, 100pF
    - Potentiometers: 50kΩ (2), 100kΩ
    - Headers and jumpers, 3-way screw terminal, DC jack
    - PCB and acrylic case
    - 9–12V DC power supply (center positive)

- Prepare board:
    - Work on the side with white printing.
- Solder small parts first:
    - Solder all resistors
    - Solder ceramic capacitors (no polarity).
    - Solder electrolytic capacitors (long lead to + mark on PCB).
- Solder larger parts:
    - Solder IC socket (align notch with PCB marking). Do not insert the XR2206 chip yet.
    - Solder J1-J2 and J3-J7 header pins.
    - Solder output terminal with openings facing outward
    - Solder DC power jack.
    - Solder three potentiometers (bend metal tabs flat).
- Trim leads:
    - Cut all leads flush on the back of the board so the case will fit.
- Insert IC:
    - Place XR2206 into the socket with notch matching socket notch.
- Power connections:
    - Connect 12V DC supply to jack.
- Set jumpers and controls:

    - J1 jumper = sine wave on SINE/TRI output, J2 jumper = triangle wave.
    - Set J3-7 to 300Hz-3kHz
    - Left knob = Amplitude (turn anticlockwise to increase).
    - Center knob = Fine frequency.
    - Right knob = Course frequency.
- Output stage:
    - Top screw terminal = GND.
    - Middle terminal = SQUARE wave output.
    - Bottom terminal = SINE/TRI wave output.

Building the Summing Box:

The passive summing box was built in order to test multiple input frequencies. The procedure was as follows:

- Gather components for the mixer:
    - Potentiometers: 10kΩ log (2)
    - Resistors: 10kΩ (2)
    - Input jacks: 1/4" TS mono (2)
    - Output jack: 1/4" TS mono (1)
    - Enclosure, hookup wire, solder
- Prepare enclosure:
    - Mount the two input jacks on the left side, and an output jack on the right side.
    - Mount the two 10kΩ pots on the top panel and label Input 1 and Input 2.
- Input 1 wiring:
    - Connect Input 1 tip to the left lug of the top pot.

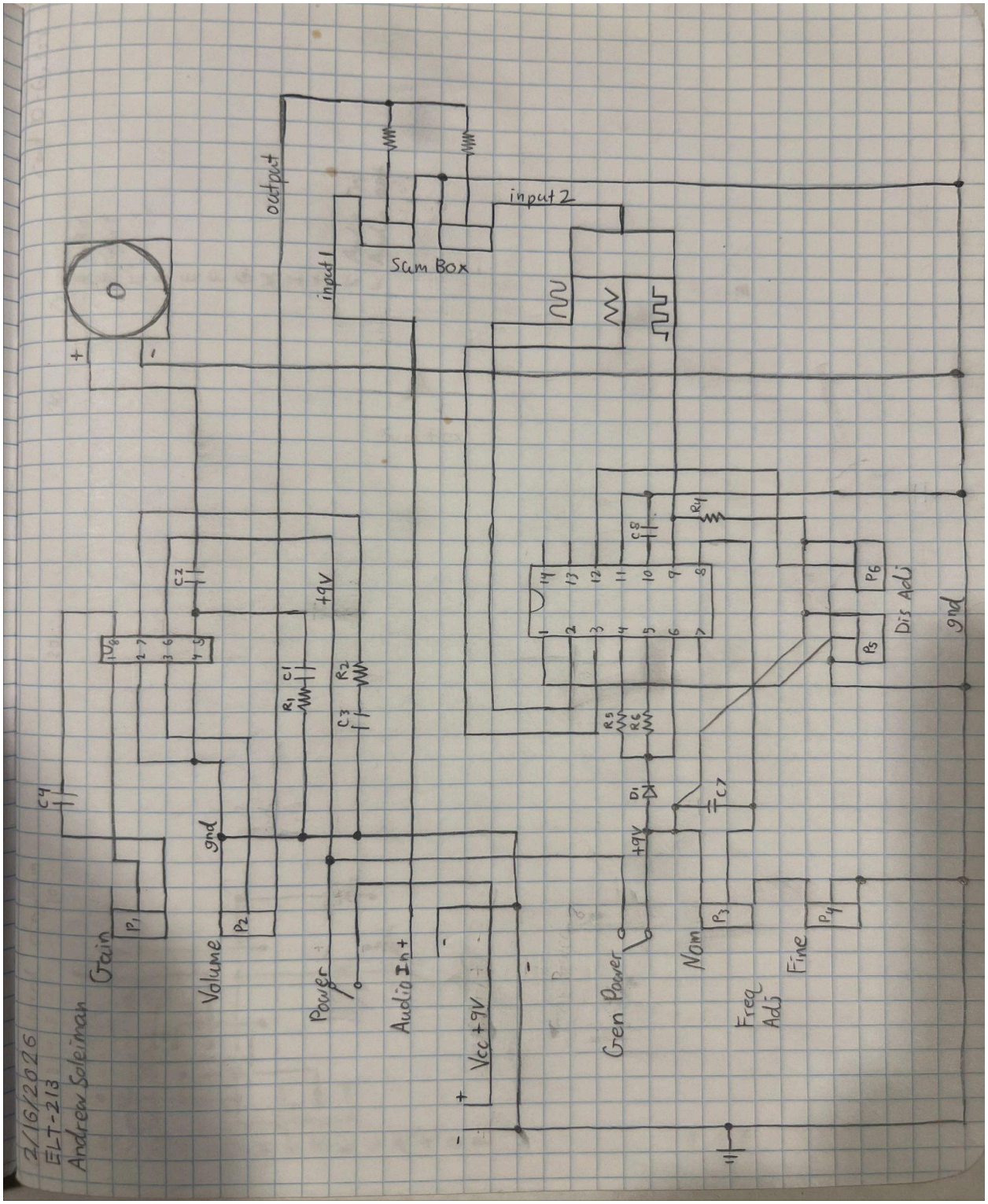
    - Connect Input 1 sleeve to ground bus.
    - Connect the right lug of the top pot to the ground bus.
    - Connect the center lug (wiper) of the top pot to the first 10kΩ resistor.
- Input 2 wiring:
    - Connect Input 2 tip to the left lug of the bottom pot.
    - Connect Input 2 sleeve to ground bus.
    - Connect the right lug of the bottom pot to the ground bus.
    - Connect the center lug (wiper) of the bottom pot to the second 10kΩ resistor.
- Summing stage:
    - Join the free ends of both 10kΩ resistors together.
    - Connect this junction to the output tip.
- Ground connections:
    - Connect Output sleeve to ground bus.
    - Connect all ground points together: input sleeves, pot right lugs, output sleeve.

Conclusion for Stage 2:

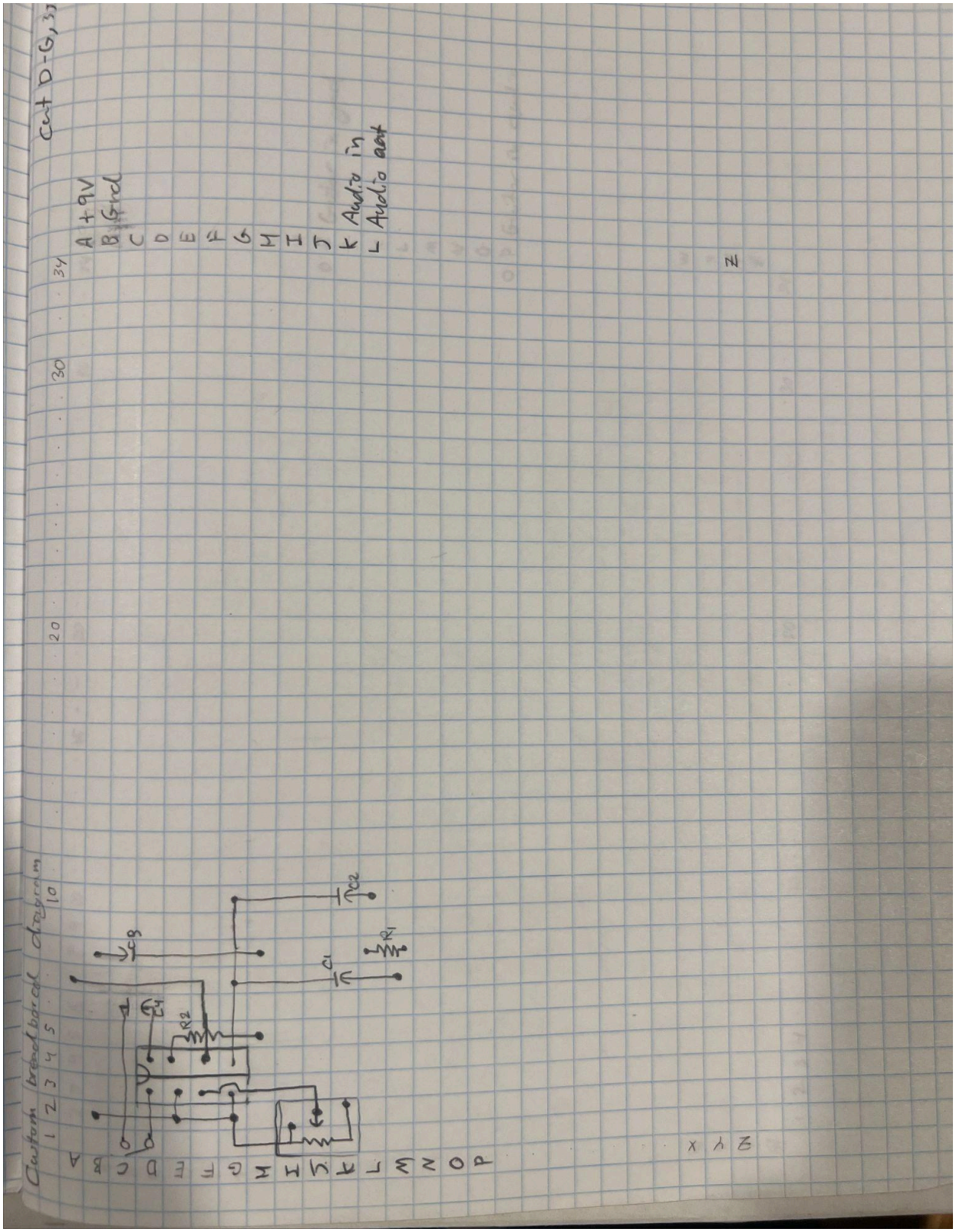
With each component built and tested using the troubleshooting steps listed in stage 1, the main circuit was functioning as expected. A new and final schematic was made for stage 3, the final assembly of the project.

Data:

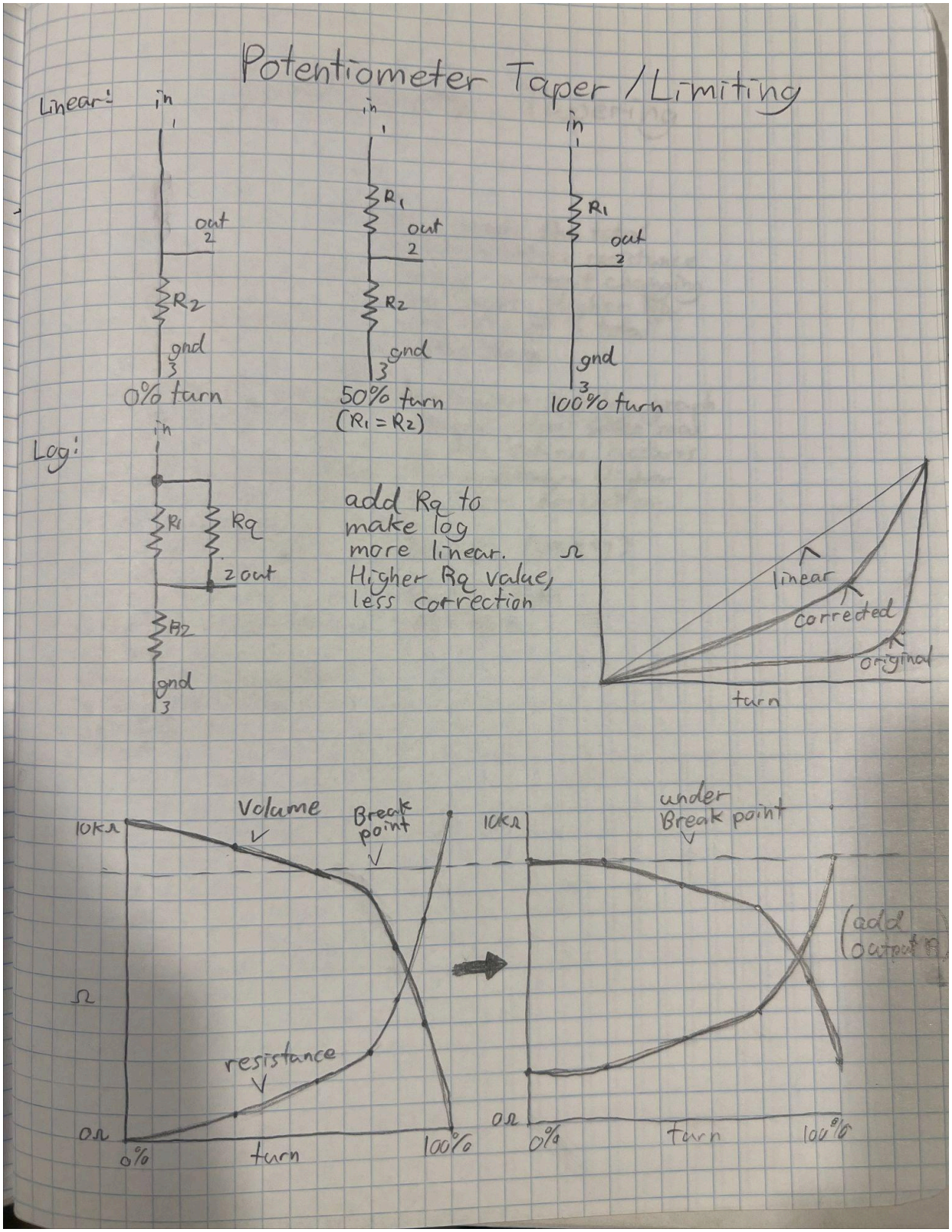
Schematic Version 2:



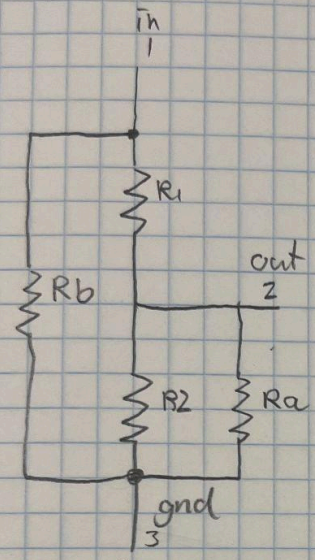
LM386 Amp Solderboard Schematics:



Potentiometer Tapering/ Limiting/ Calculations:



Potentiometer Tapering



R b lowers the total potentiometers resistance value without changing the taper. Higher Rb, the less of a total value drop.

Ra makes the log graph steeper, ideal when max or min volume occures to suddenly. Higher Ra, less correction.

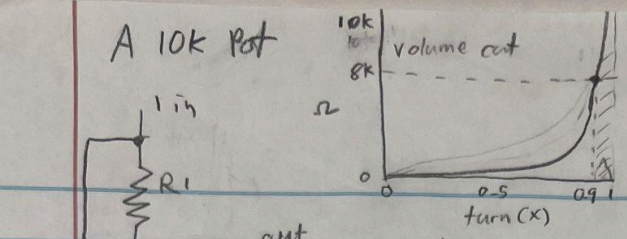
$Ra = R2 \cdot (0<x<1)$

$R1 = (1-x)R1$

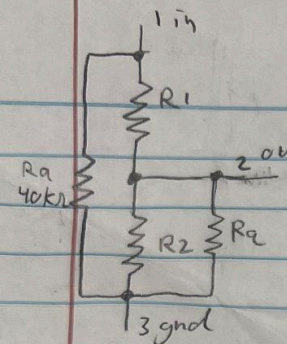
$R2 = R2\,x$

x is the amount of turn in a linear potentiometer. 0 being 0%, and 1 being 100%

A 10k Pot



Volume = $\frac{1}{R_1}$ => Volume $\propto R_1$



$R_T = \frac{1}{1/R_1 + 1/R_a}$

$8k\Omega = \frac{1}{1/10k + 1/R_a} \Rightarrow \frac{1}{8k\Omega} = \frac{1}{10k\Omega} + \frac{1}{R_a}$

$R_a = \frac{1}{1/8k - 1/10k} = \boxed{40k\Omega}$

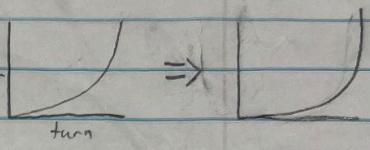
$R_a = 40k\Omega$

$R_2 = (1-x)10k\Omega$

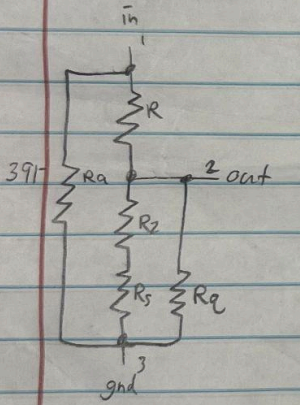
$R_1 = R_1 \cdot x = 10k\Omega \cdot x$

$R_q = R_T \cdot [0,1]_\Omega$

$R_q \propto$ correction



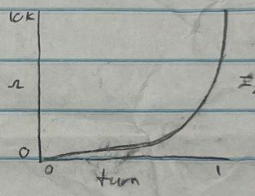
=>



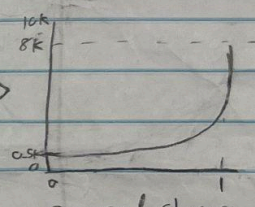
$R_a = 39k\Omega$

$R_3 = 470\Omega$

$R_q = 2.2k\Omega$

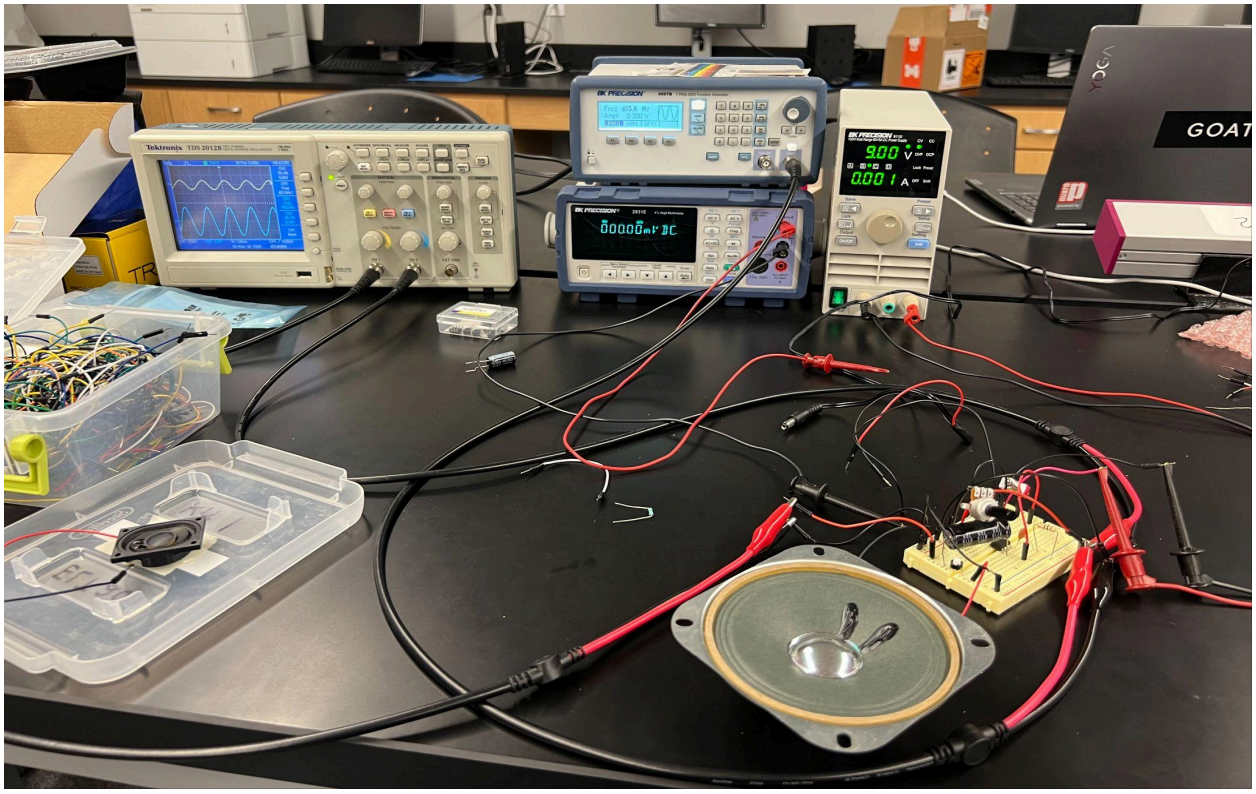


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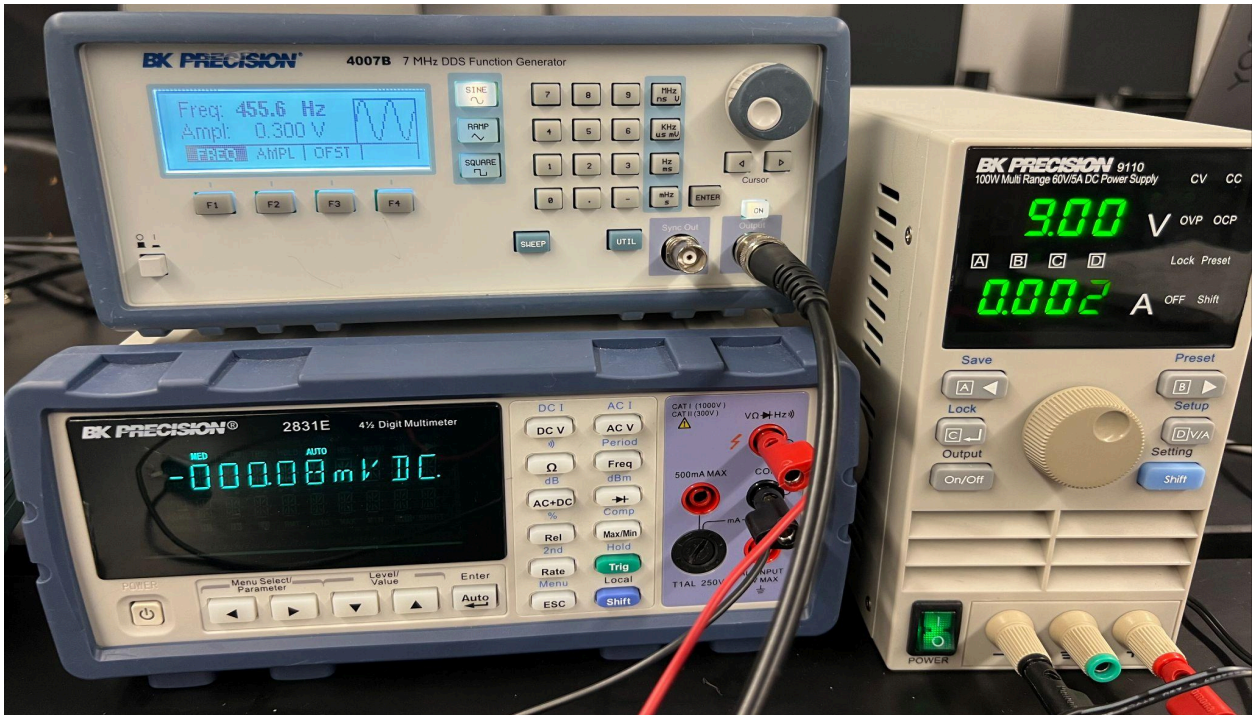


capped, steeper

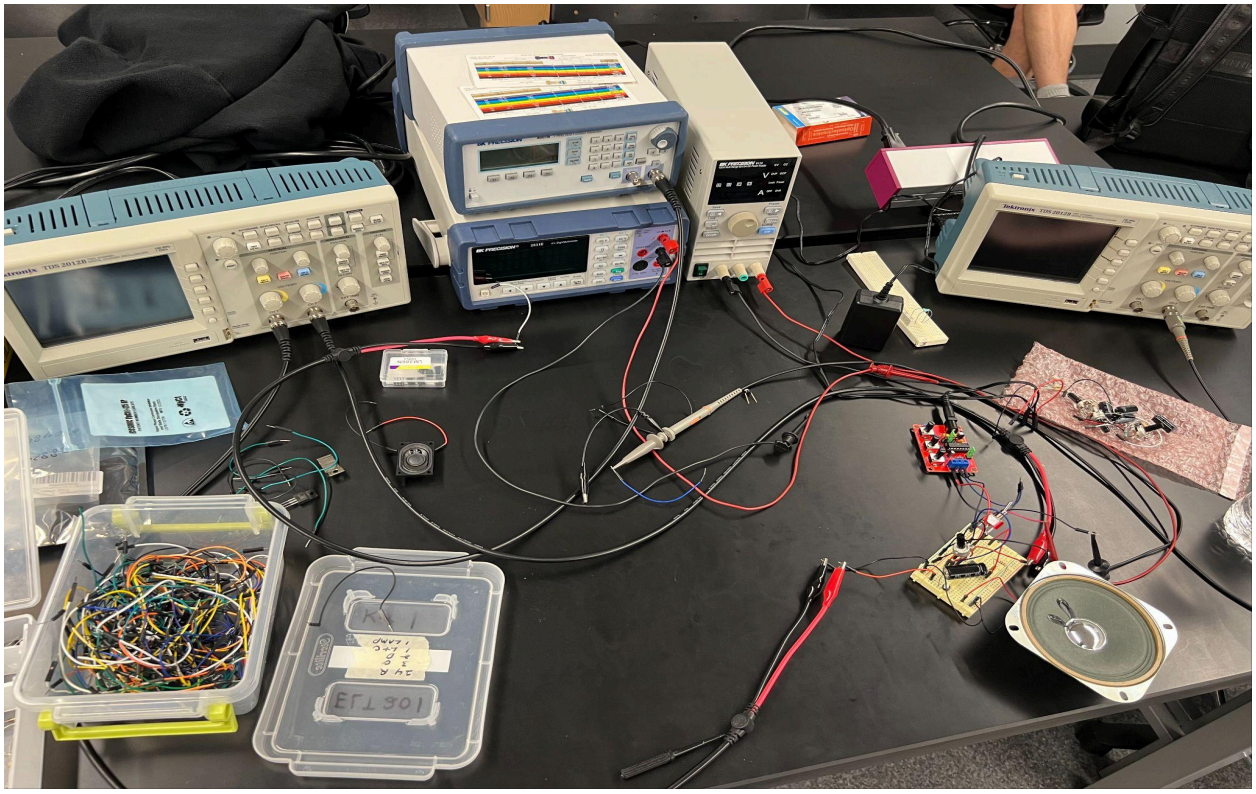
LM386 Amp Test Build:

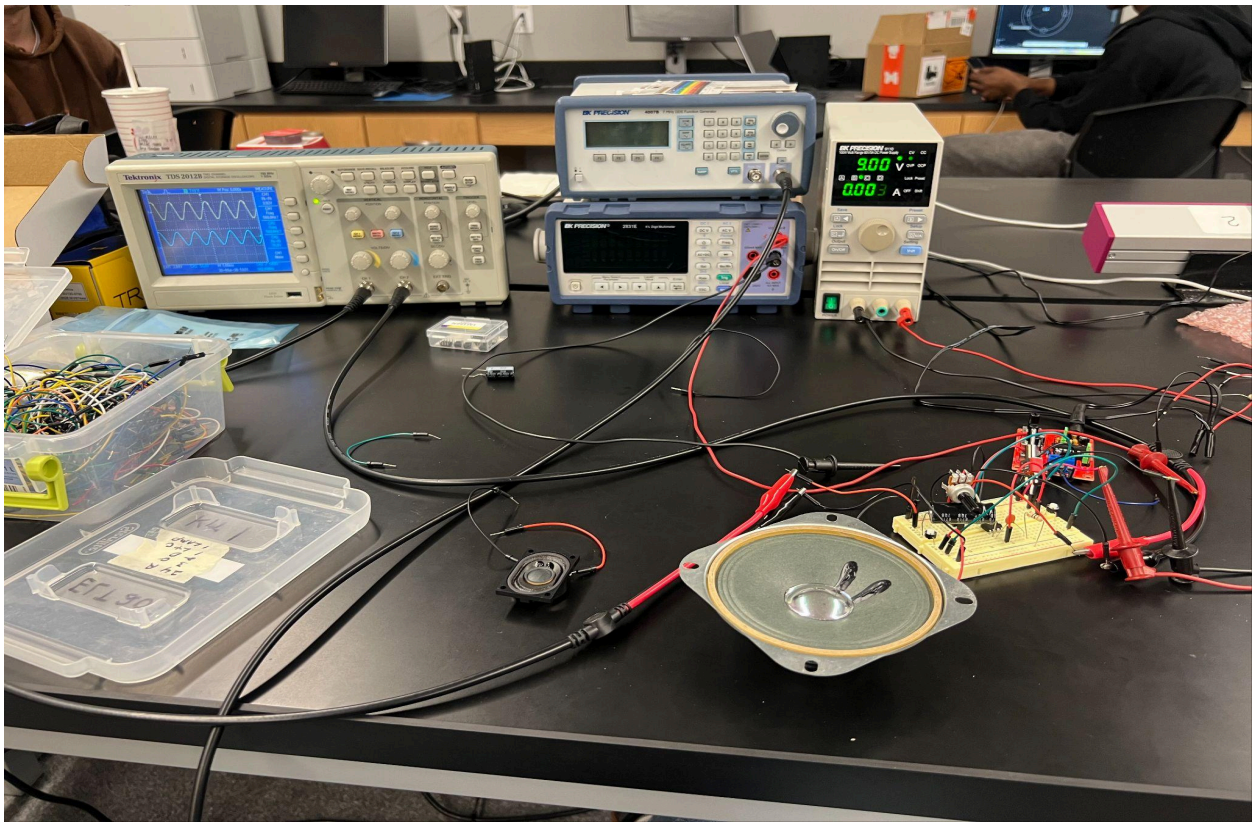


Results from Test Build:

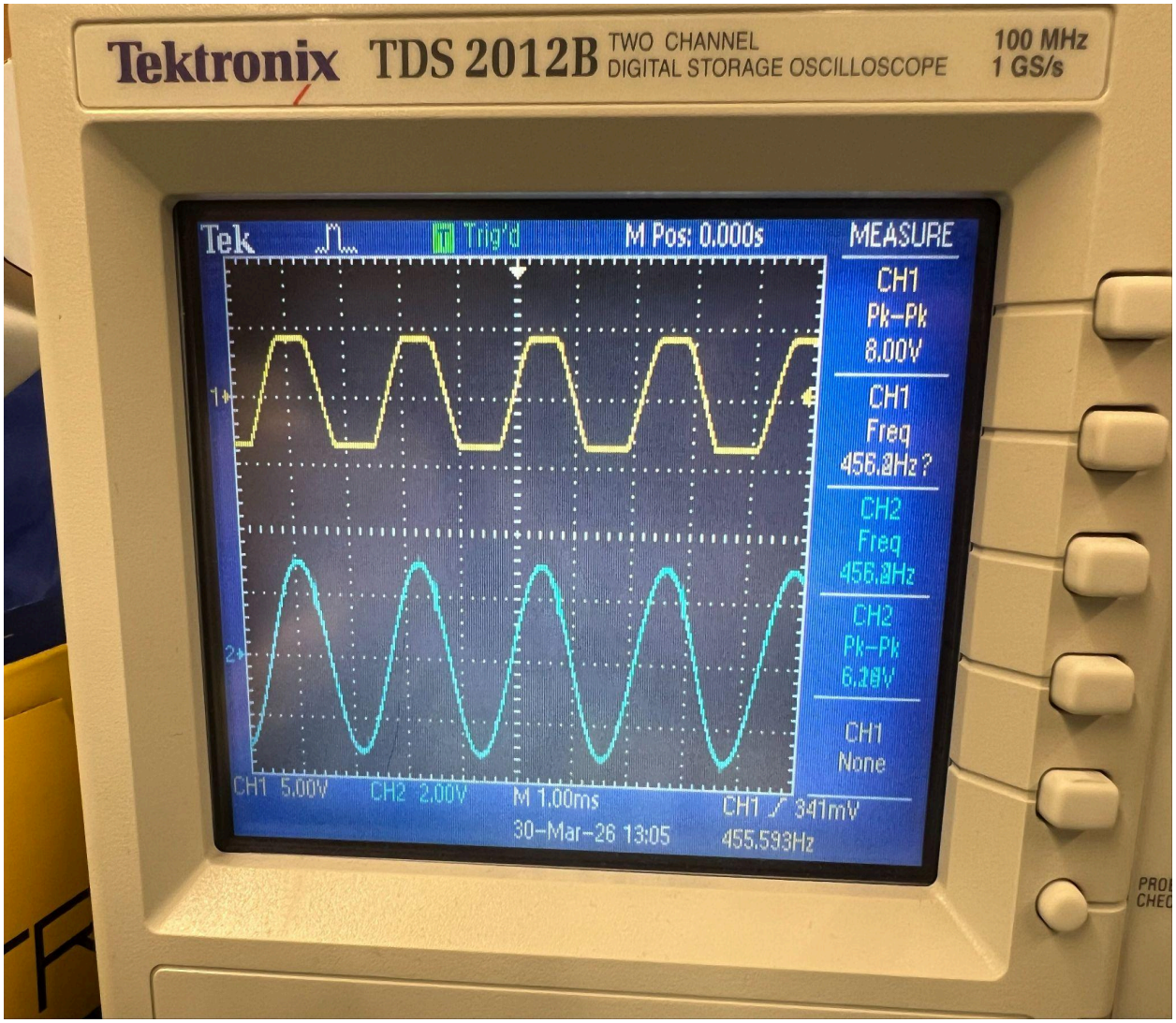


LM386 Amp with XR2206 Gen as Input:

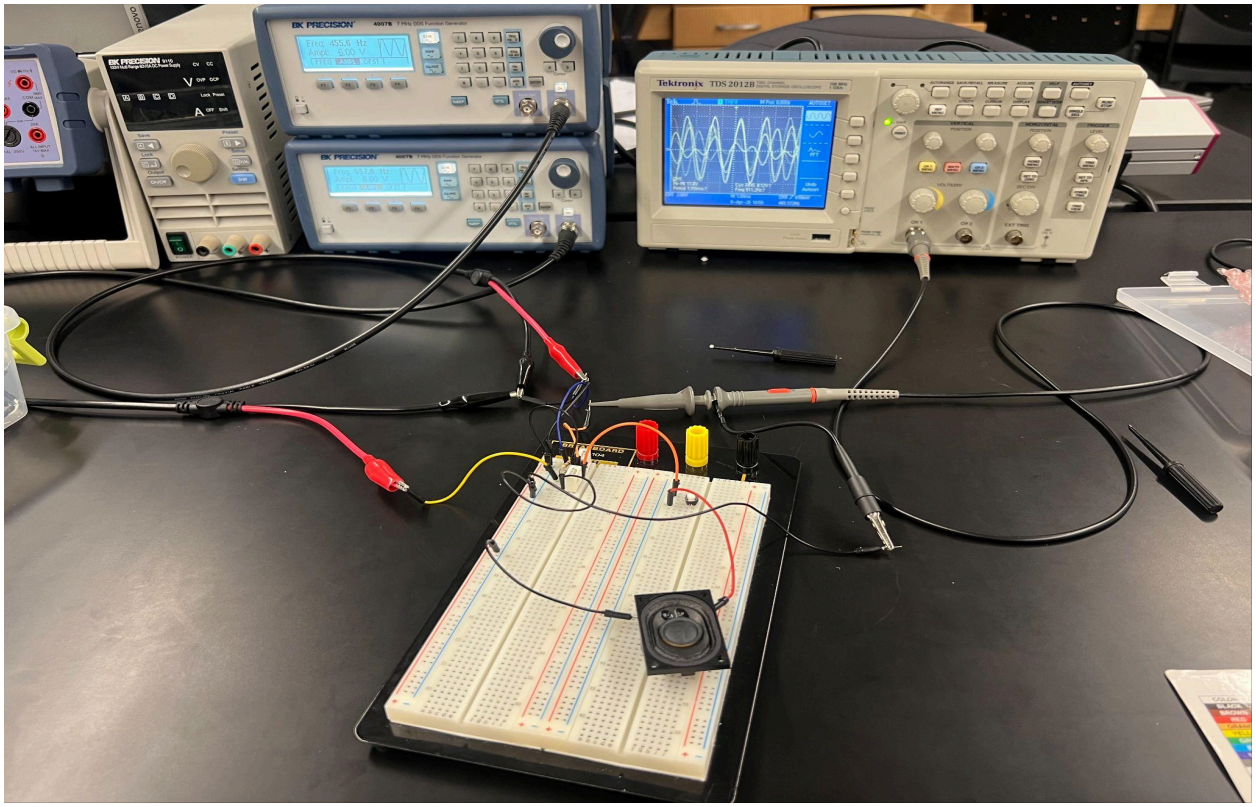


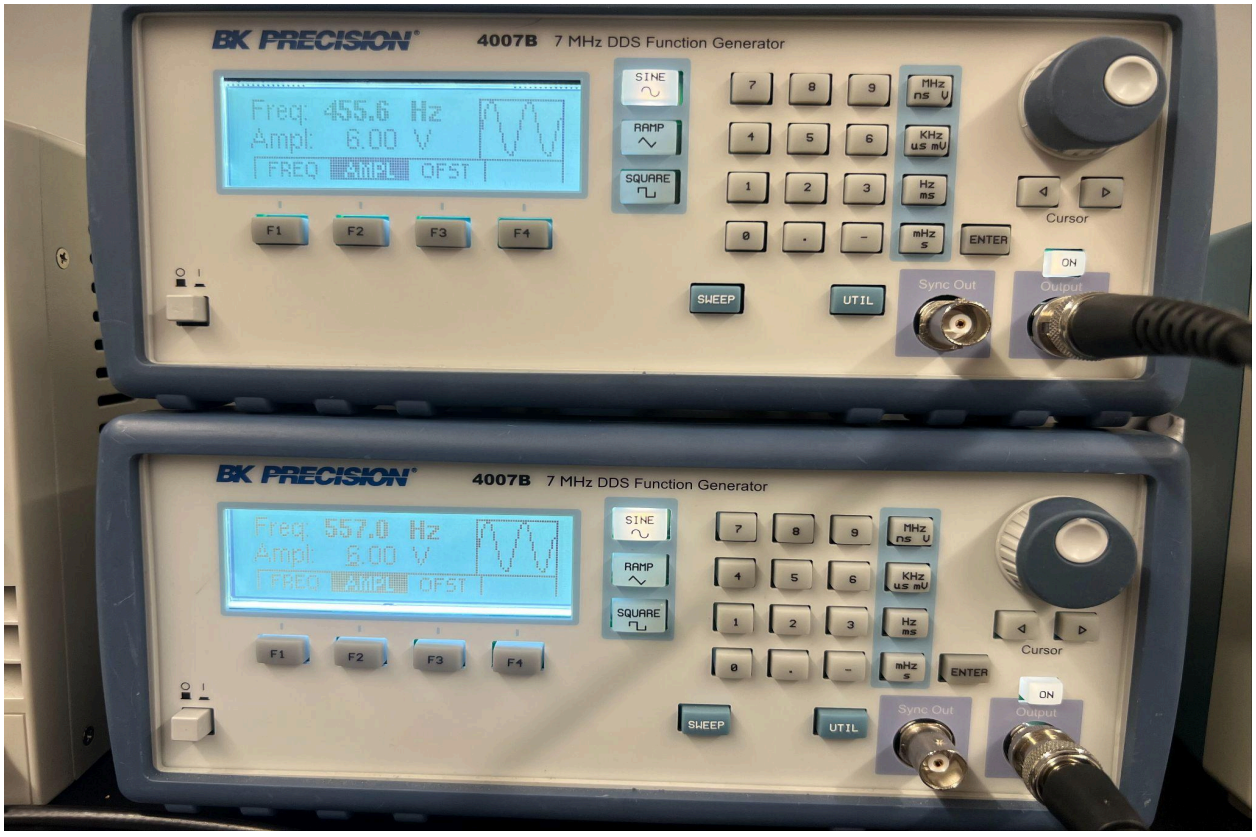


Results from LM386 Amp and XR2206 Input:

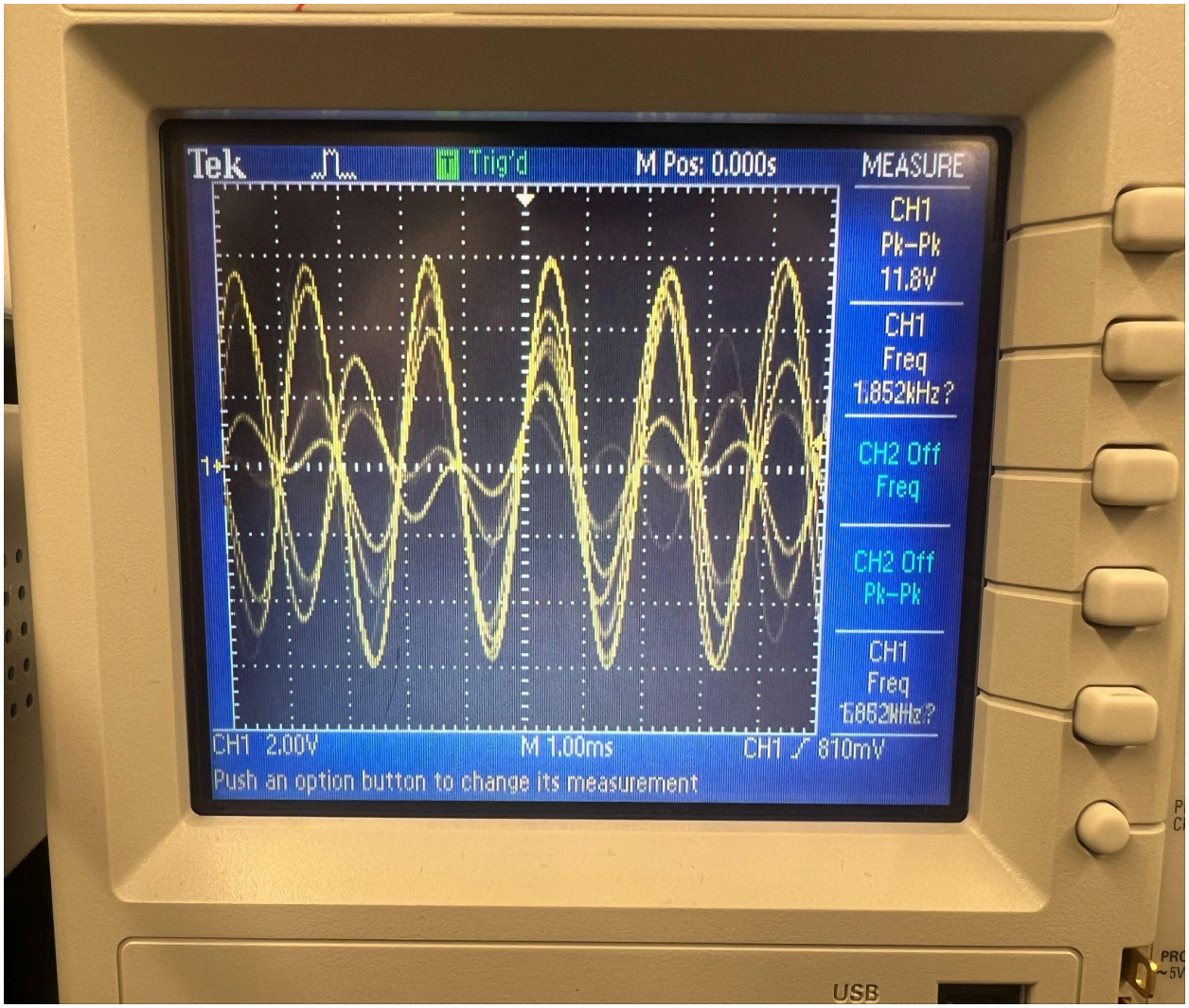


Summing Box Test Build 1:

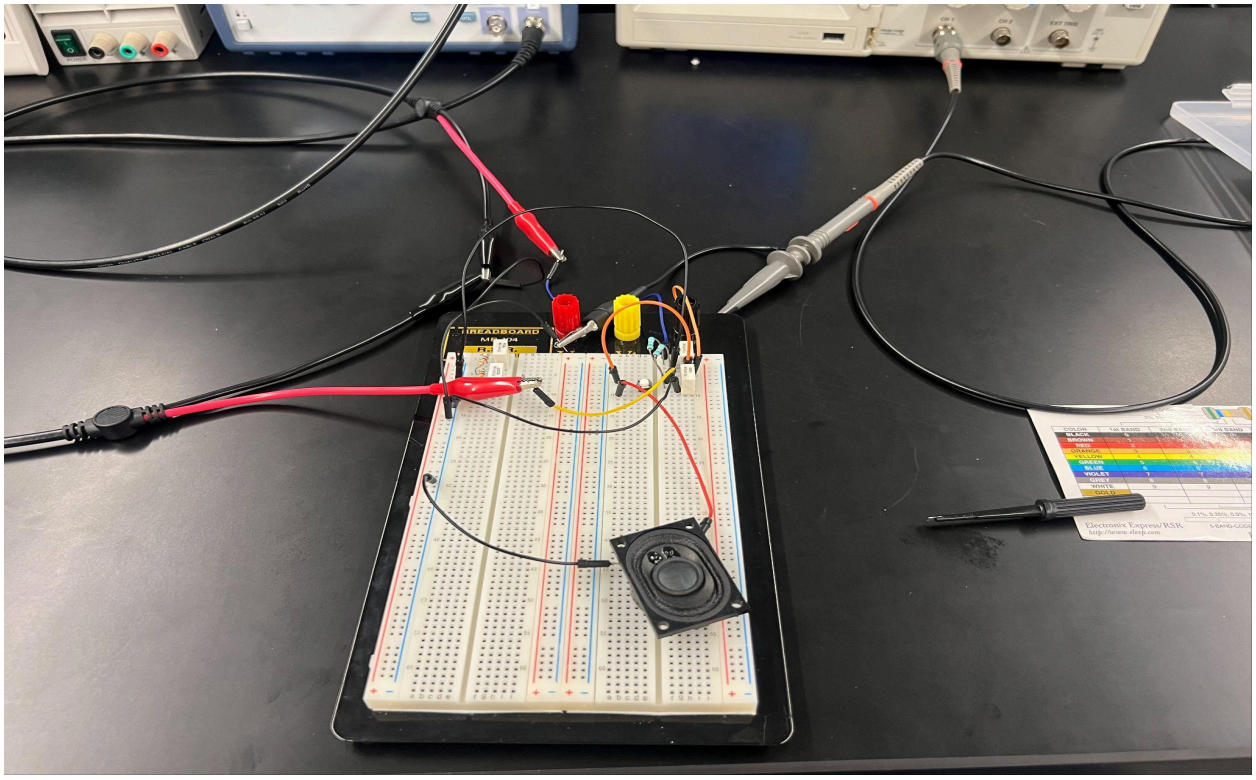




Results From Summing Box Test Build 1:

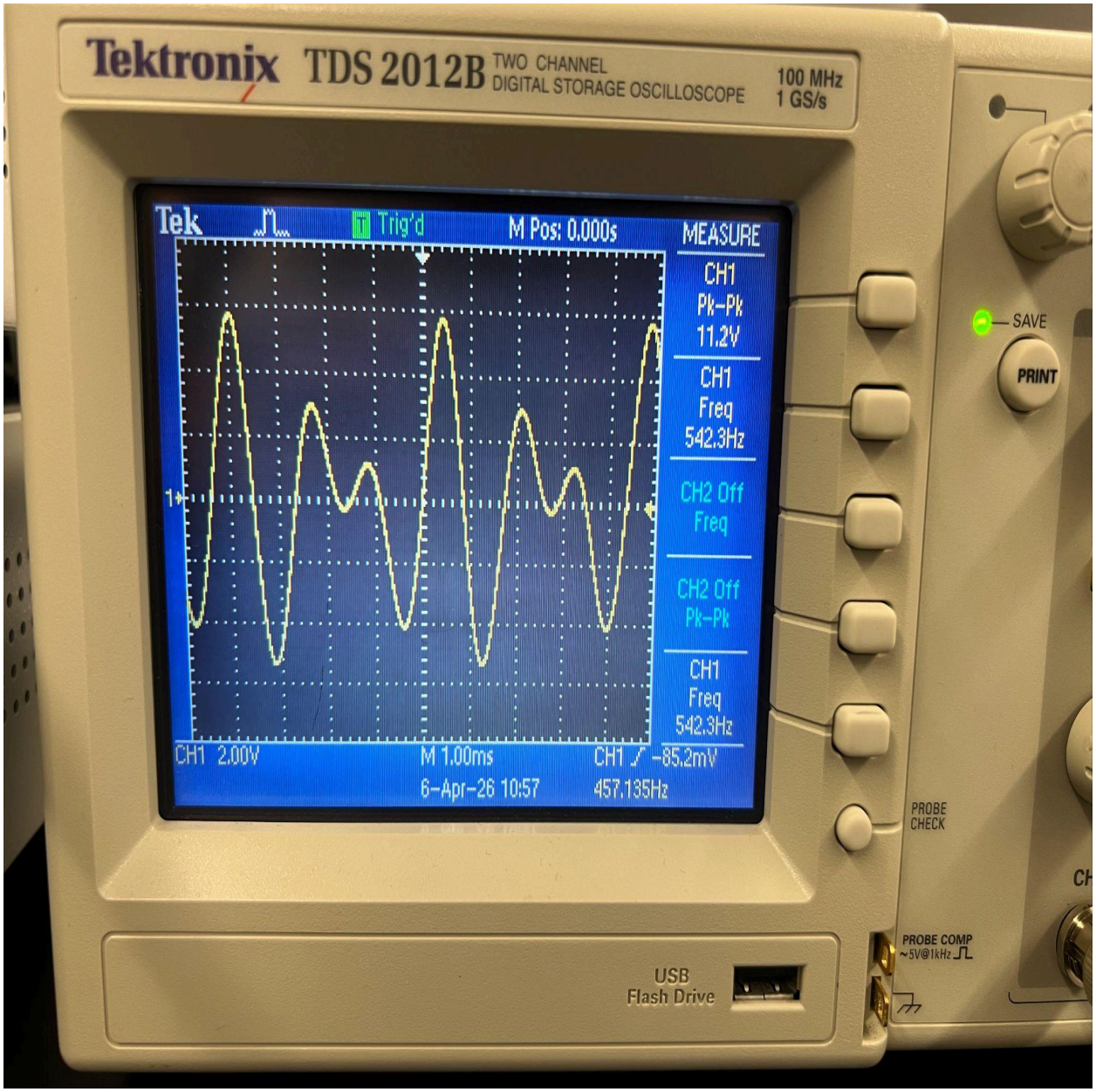


Summing Box Test Build 2 (Larger Resistance Values):

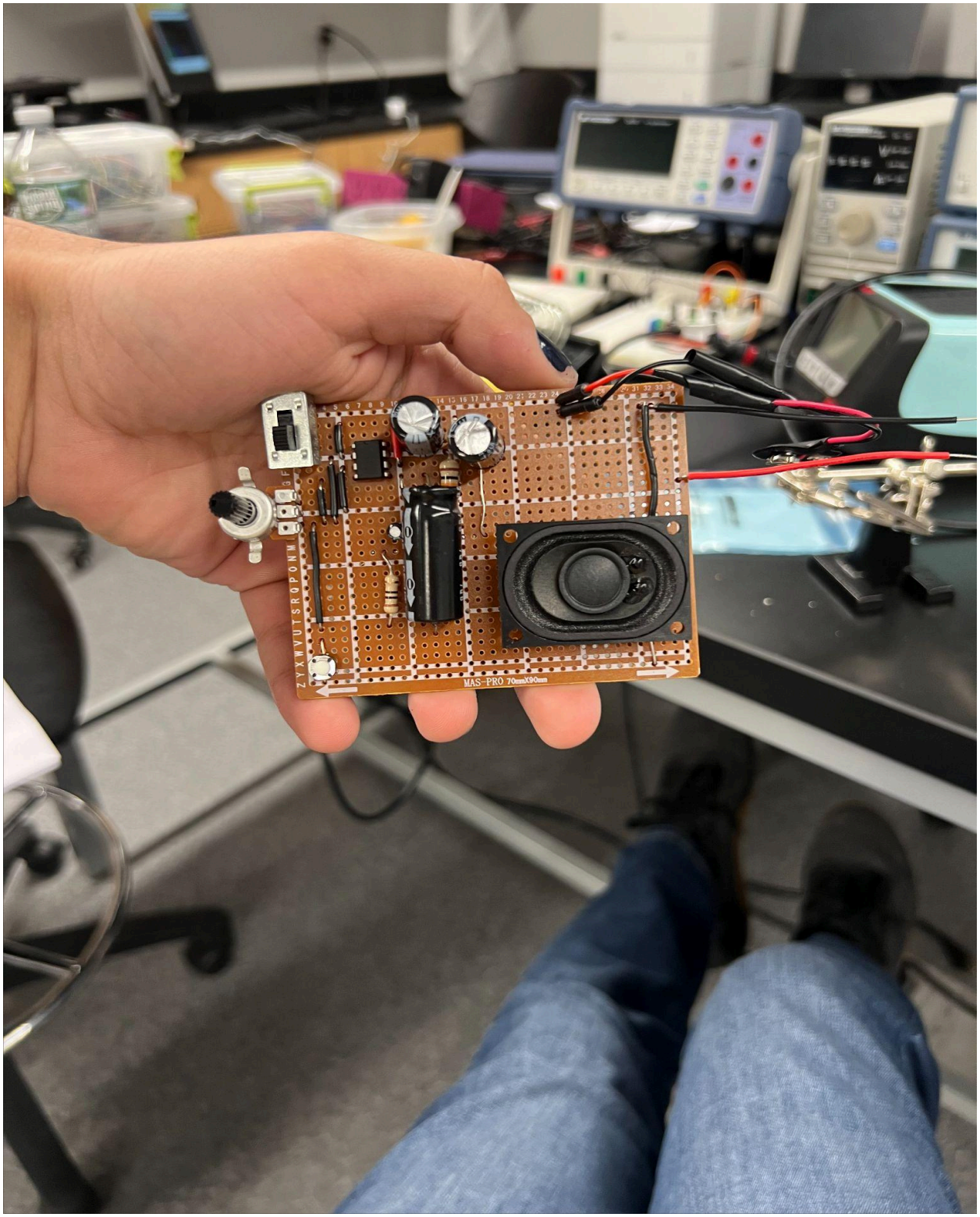




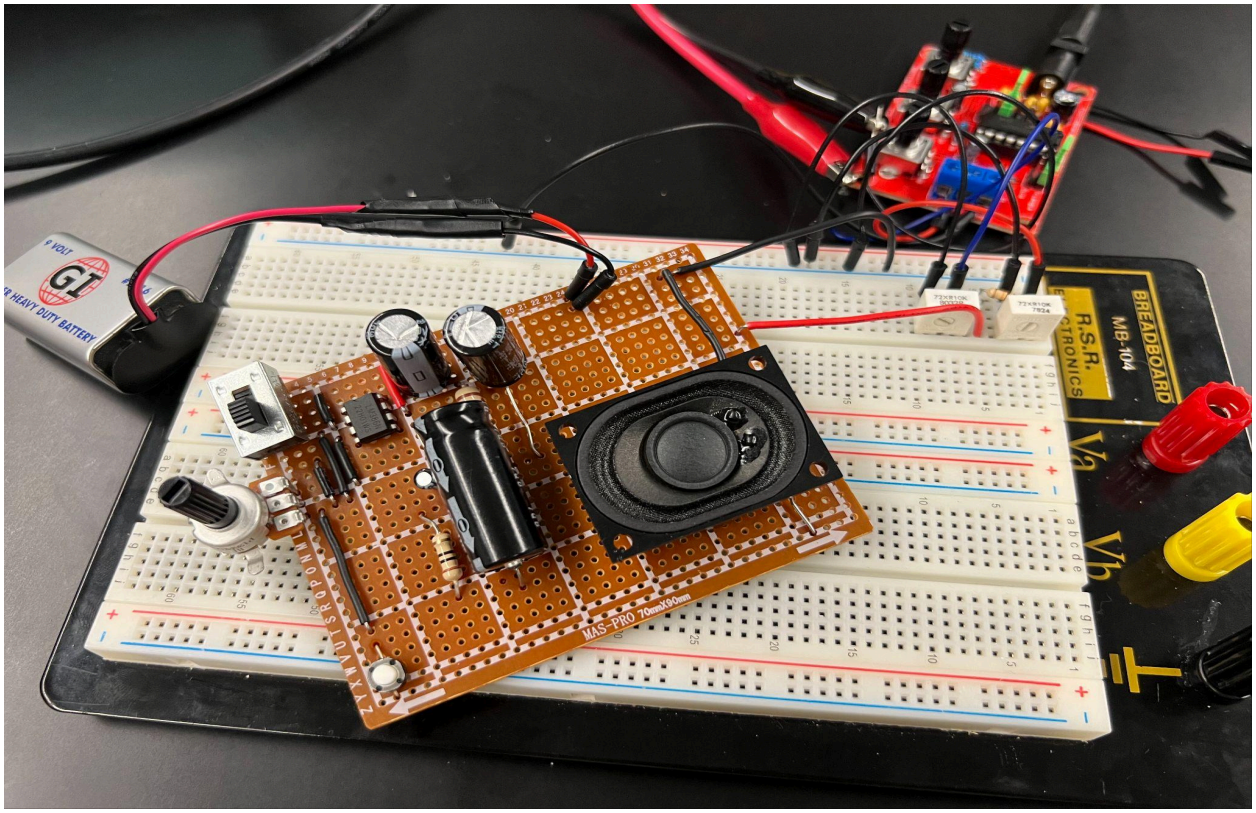
Results From Summing Box Test Build 2:

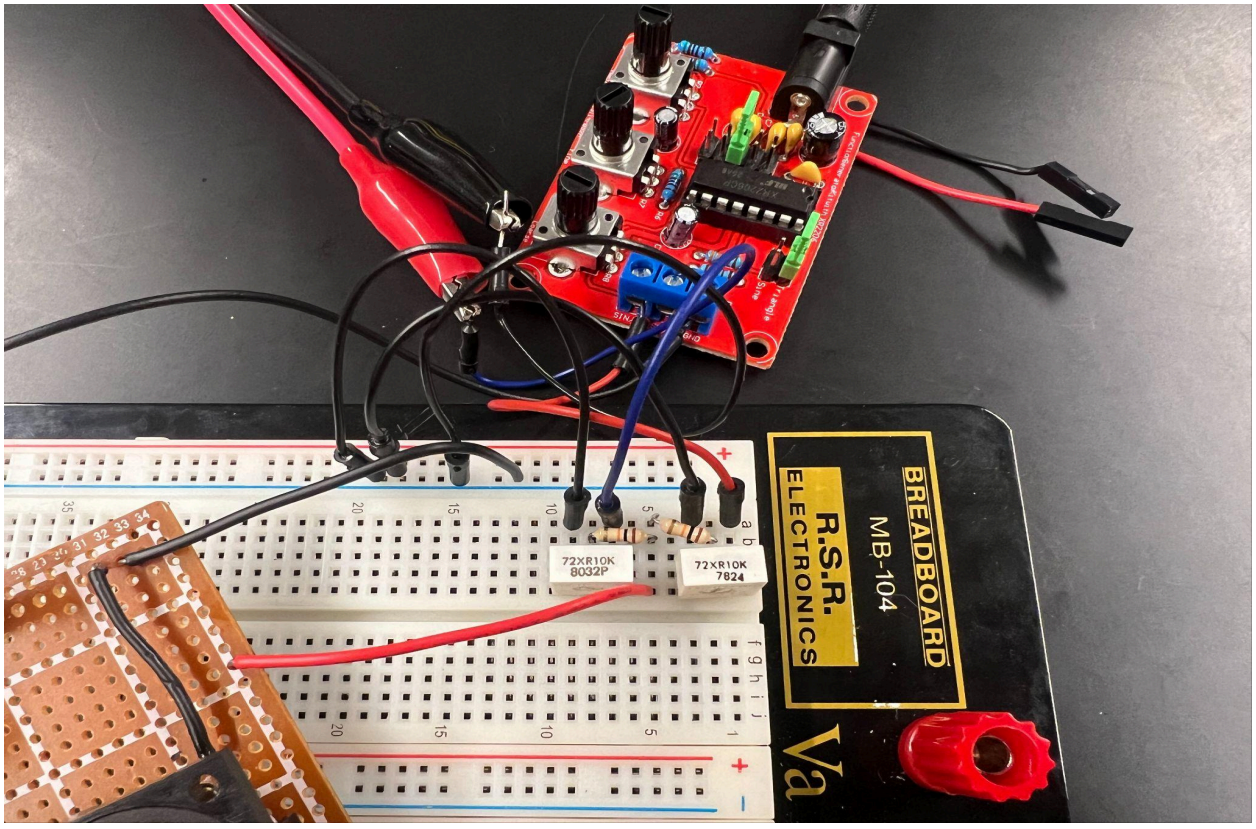


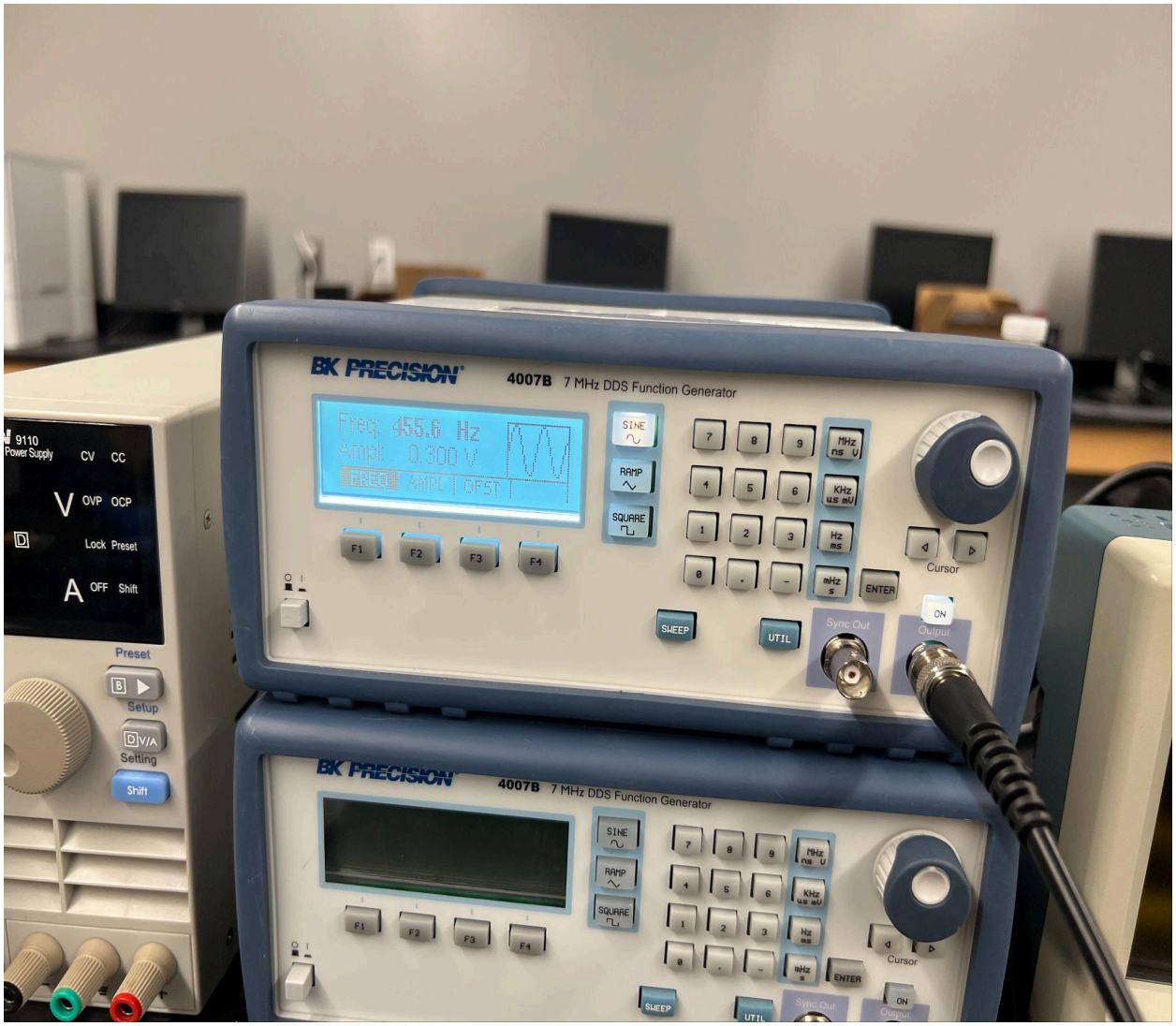
LM386 Solderboard Build 1:



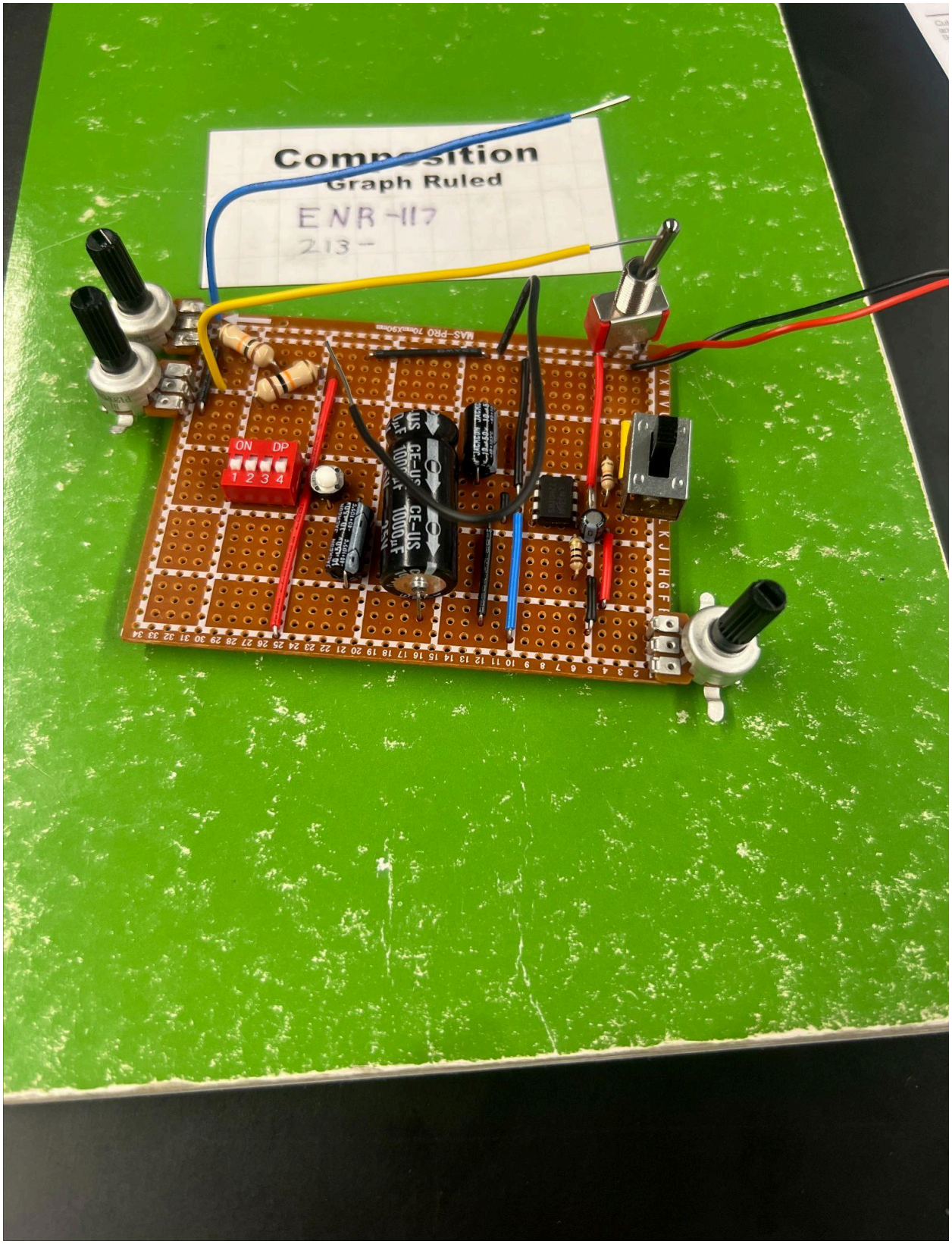
LM386 Solderboard 1 with Dual Input:



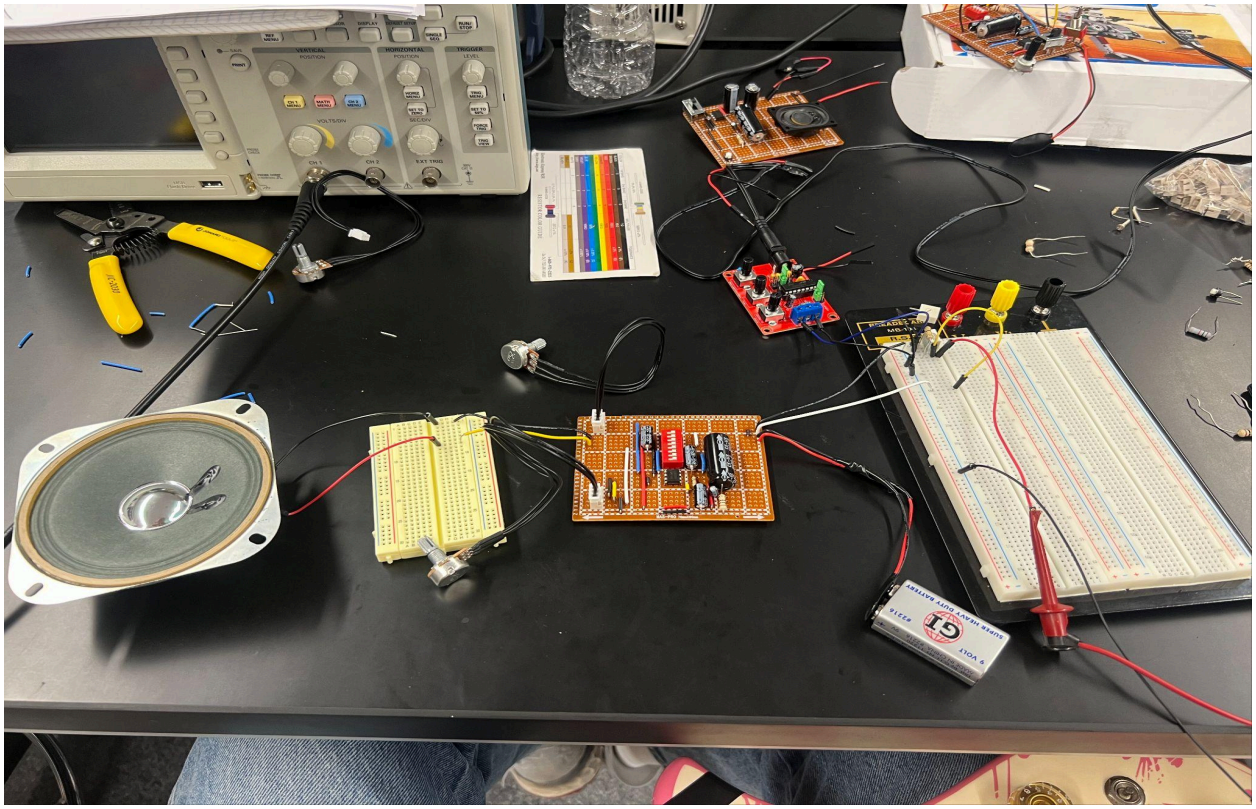



BK PRECISION
4007B 7 MHz DDS Function Generator
Freq 455.6 Hz
Ampl 0.300 V
SINE
RAMP
SQUARE
F1
F2
F3
F4
SWEEP
UTIL
Sync Out
Output
ENTER
Cursor
9110
Power Supply
CV CC
OVP OCP
Lock Preset
OFF Shift
Preset
Setup
Setting
Shift

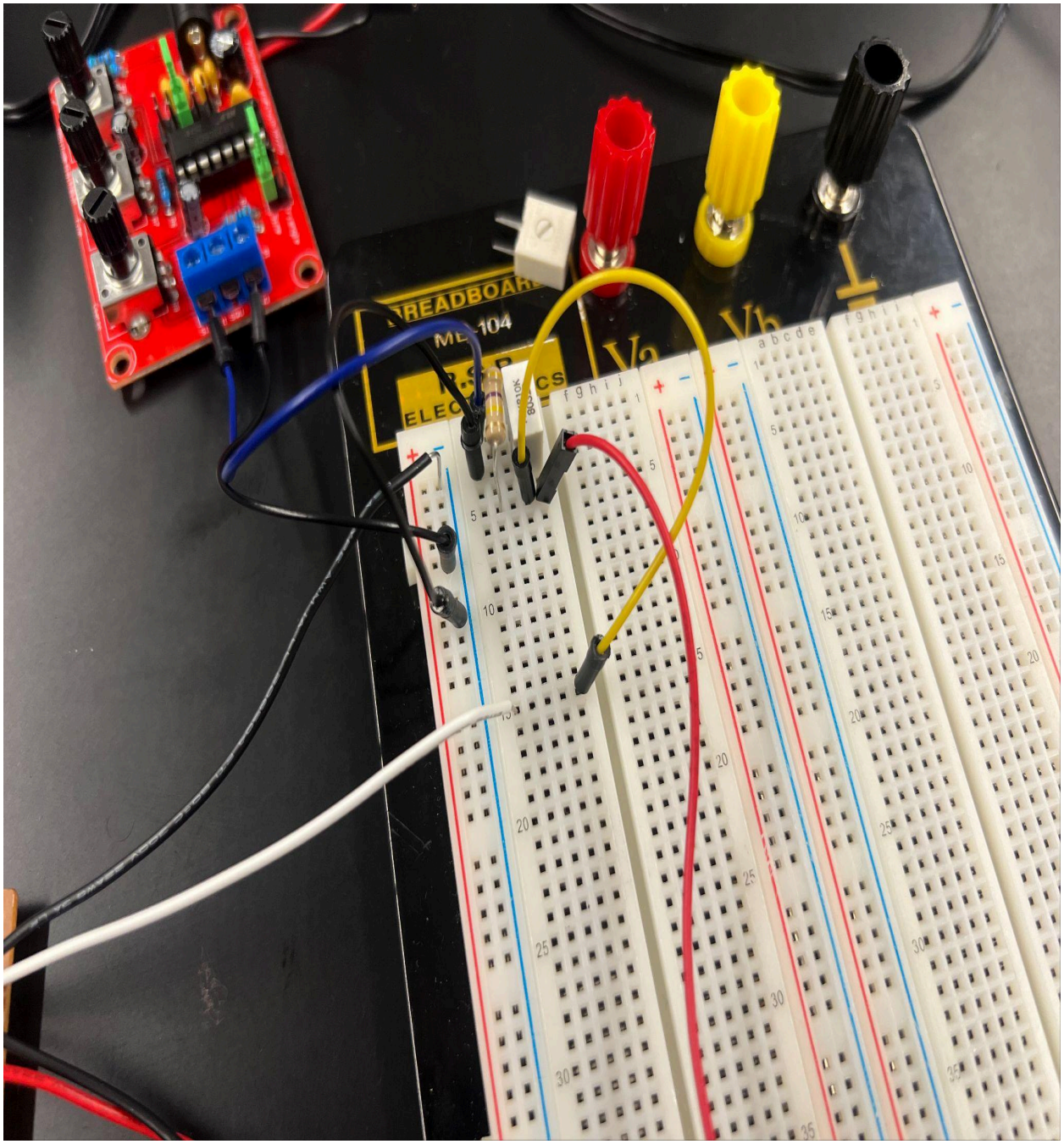
LM386 Solderboard 2 (Faulty):



Solderboard 3 with XR2206 and Guitar Input:







Stage 3 Intro:

With all modules soldered and functioning as expected, the final assembly of the project can start.

The primary purpose of this stage is to:

- Fine tune the circuit using resistors, capacitors, and potentiometers
- Create a 3d housing/ mounting board for final assembly
- Create a wiring diagram
    - Schematics
    - Cable management
    - Project board placement
- Add final modifications to the project
- Fully test the project
- Make a final parts list
- Create building instructions for complete project

Creating the Housing:

With all the circuit components working as intended, a housing was needed to hold everything together. The initial idea was to 3d print a box, big enough to contain all the wiring, switches, and circuit boards. However, a 3d printed box would require tight and condensed wiring, as space would be limited, making assembly and maintenance extremely difficult. A 3d printed housing box would also be visually unappealing as 3d printers leave rough grooves all throughout the object. To make a 3d printed box visually appealing it would require sanding, priming, and painting, all of which is time consuming and expensive. With that in mind, an open faced design was selected, meaning all of the wiring and components would be mounted onto a 3d printed plate. Since the face of the plate would be covered in neatly managed wires and circuitry, the 3d printed grooves would be hidden. This would also mean any modifications or maintenance needed to be done would be effortless. A 11in x 8in x 0.25in plate was printed for the project. Mounting holes were then drilled and all switches, inputs, and outputs were glued to the edges of the board.

Modifications:

- **Dual Note Floor Pedal:**
    - Although the drone note guitar amplifier was working as intended, it did not have the ability to change the drone note while playing. The solution for this was to have a NC (normally closed) floor pedal connected to the guitar amp. When the pedal was pressed, the drone note would change. The initial design for this was to use an elaborate array of transistors acting like logic gates, allowing a different drone note to play only when the pedal was pressed. This design required 2 XR2206 oscillators and a bulky transistor logic board. A new design implemented a potentiometer in parallel with the “Course” tuning potentiometer on the XR2206 board. When the pedal was closed, a larger resistance value would lower the drone note, resulting in a second tunable drone note.
- **Headphones/ Speaker Output:**
    - While building this project, the LM386 amplifier chip was discovered to be better suited for headphones rather than speakers. Although the speaker gave satisfactory results in terms of volume and quality, testing the circuit with headphones gave a much richer and cleaner output. An audio output toggle switch was installed to allow either a speaker output or a headphones output. This modification was included in the wiring schematics.
- **Alternative Amplifier Design:**
    - This project uses a common LM386 amplifier circuit. This circuit has gain control but little frequency filtering and no tone control. The Ruby Amp design was considered for this project. The Ruby Amp was also based on the LM386 IC but had an input buffer to retain high-frequency detail, resulting in higher quality. However, this design was more complicated and also included a transistor. The standard LM386 amp circuit was good enough for the project and was preferred over the Ruby Amp.

Creating a Wiring Schematic:

All components were laid on the 3d printed plate for positioning. Once everything was glued down, a cable management diagram was made to route the wires. The new wiring diagram was then used to wire the circuits together along with the added modifications.

Final Assembly Instructions:

Now that all of the circuits and modifications have been constructed and tested, the project is ready for the final assembly.

The procedure was as follows:

- **Glue Down Components to Edge of Plate:**
    - Guitar jack
    - +12 DC in
    - XR2206 on switch
    - LM386 on switch
    - speaker/ headphones switch
    - headphones out jack
    - floor pedal in
- **Glue Down Remaining Components to Middle of Plate:**
    - speaker stand offs
    - Volume, gain, mix, and 2nd drone note potentiometers
- **Secure Boards and Speakers:**
    - XR2206 oscillator board
    - LM386 amplifier board
    - Speaker
    - +9 volt battery holder

- **Wiring:**
    - Guitar In Jack to LM386 input 1
    - +12 DC In to XR2206 on switch
    - XR2206 on switch to XR2206 oscillator
    - +9 volt battery to LM386 on switch
    - LM386 on switch to LM386 amp
    - XR2206 output to LM386 input 2
    - LM386 out to output toggle switch
    - Output toggle switch to speaker
    - output toggle switch to headphone out jack
    - Volume potentiometer to LM386 amp
    - Gain potentiometer to LM386 amp
    - Mix potentiometer to LM386 summing circuit
    - XR2206 “course” potentiometer to floor pedal in
    - Floor pedal in to 2nd drone note potentiometer
    - 2nd drone note potentiometer to XR2206 “course” potentiometer (in parallel)
- **Secure wiring:**
    - Use zip ties through routing holes to secure wires

Final Testing:

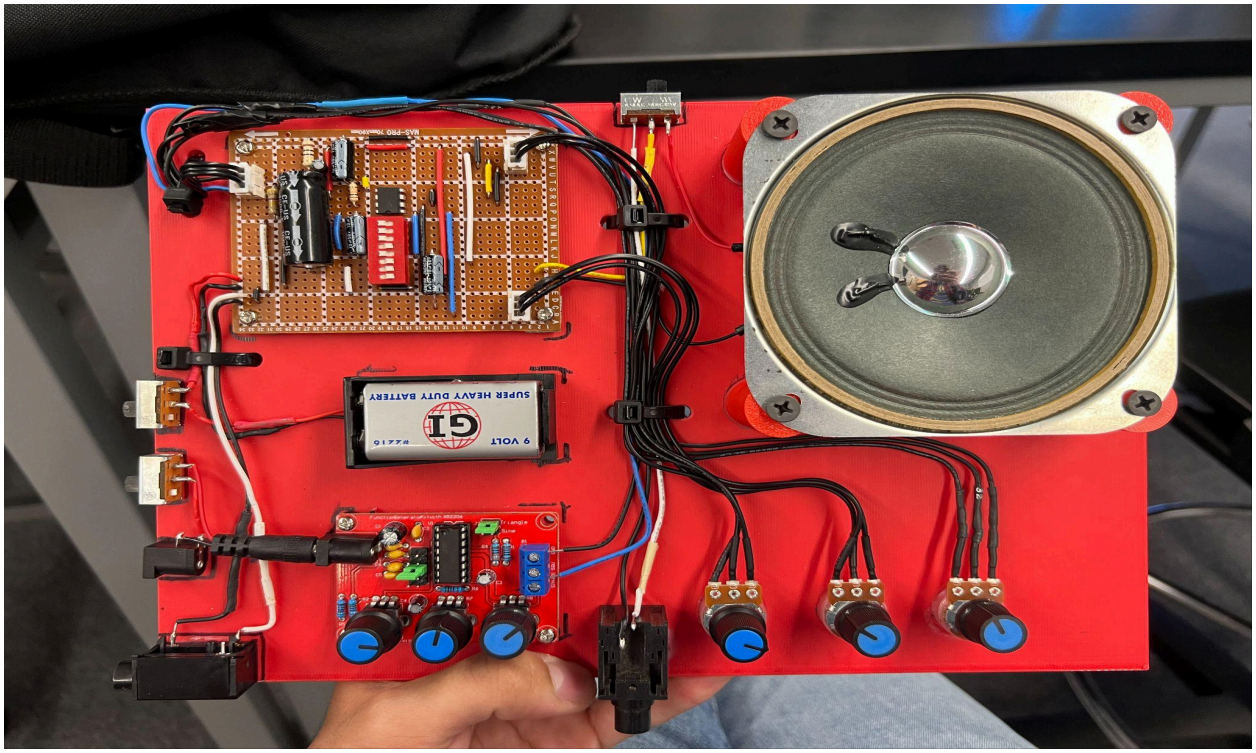
With everything assembled, the completed project was tested from start to finish. A +12 volt DC power supply was plugged into the jack on the side and both "on" switches were turned on for the amp and oscillator. The volume knobs were adjusted slowly until the drone note was audible through the speaker. The frequency knobs were then adjusted to test the drone note adjustability across the full range. Both "on" switches were turned off and a guitar was plugged into the guitar "in" jack on the front panel. The amp was turned on again and the volume and gain knobs were adjusted until a clear and audible sound was heard with no distortion. The floor pedal was connected to the input and tested for switching. Everything was working as intended and matched the design goals.

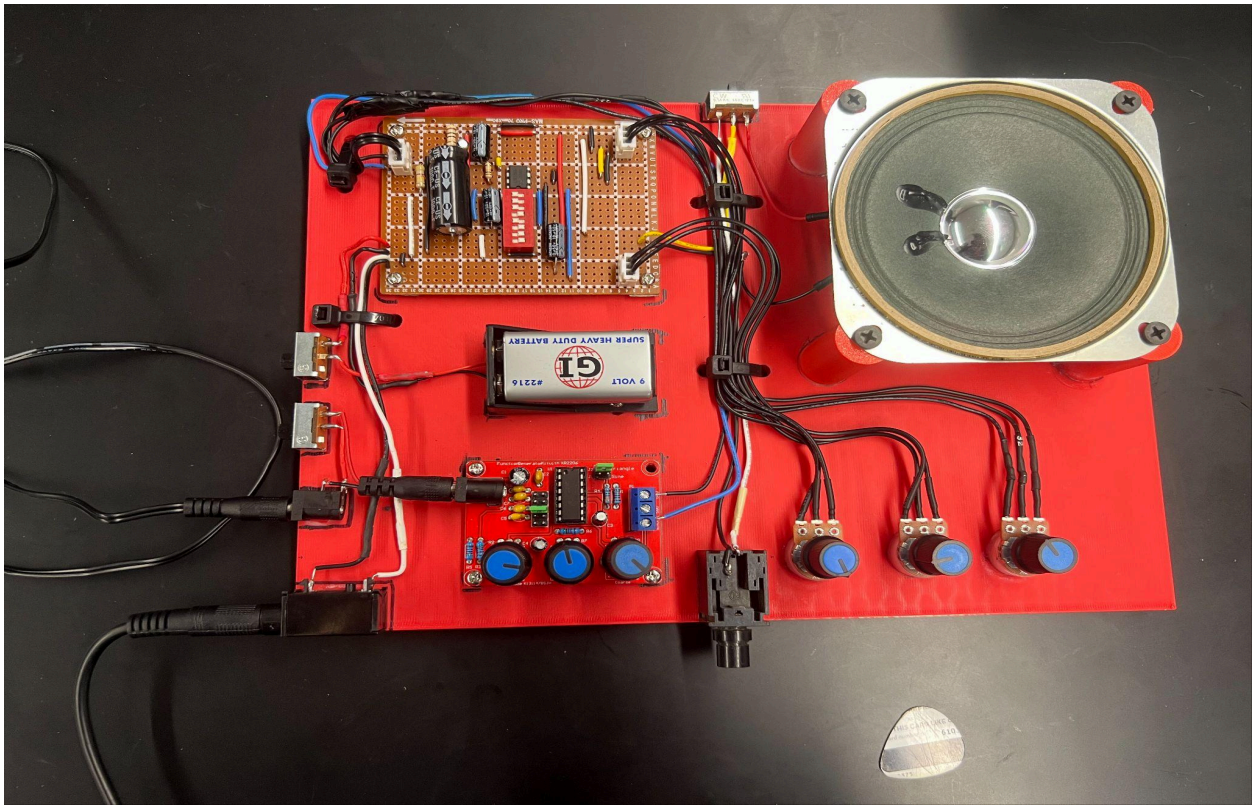
New Parts List:

- 3 x 3 pin switch
- 1 x DC jack
- 1 x +12 volt DC power supply
- 2 x ¼ in audio jack
- 1 x 8 ohm, 6 watt speaker
- 1 x +9 volt battery
- 1 x +9 volt battery holder
- 1 x 10k log potentiometer
- 2 x 10k linear potentiometer
- 1 x 250k linear potentiometer
- 1 x 100k ohm linear potentiometer
- 2 x 50k ohm linear potentiometer
- 1 x 2 pin connector
- 1 x 2 pin push button
- 1 x LM386 IC
- 1 x 470k ohm resistor
- 1 x 10 ohm resistor
- 1 x 10k ohm resistor
- 1 x 1k ohm resistor
- 1 x 5.1k ohm resistor
- 1 x 330 ohm resistor
- 4 x 10uF capacitor
- 1 x 0.1uF capacitor
- 1 x 1000uF capacitor
- 1 x 100uF capacitor
- 1 x 104 capacitor
- 1 x 105 capacitor
- 1 x 473 capacitor
- 1 x 222 capacitor
- 1 x 101 capacitor
- 1 x XR2206 IC
- 2 x 2 pin jumper cap
- 1 x 3 signal wire terminal

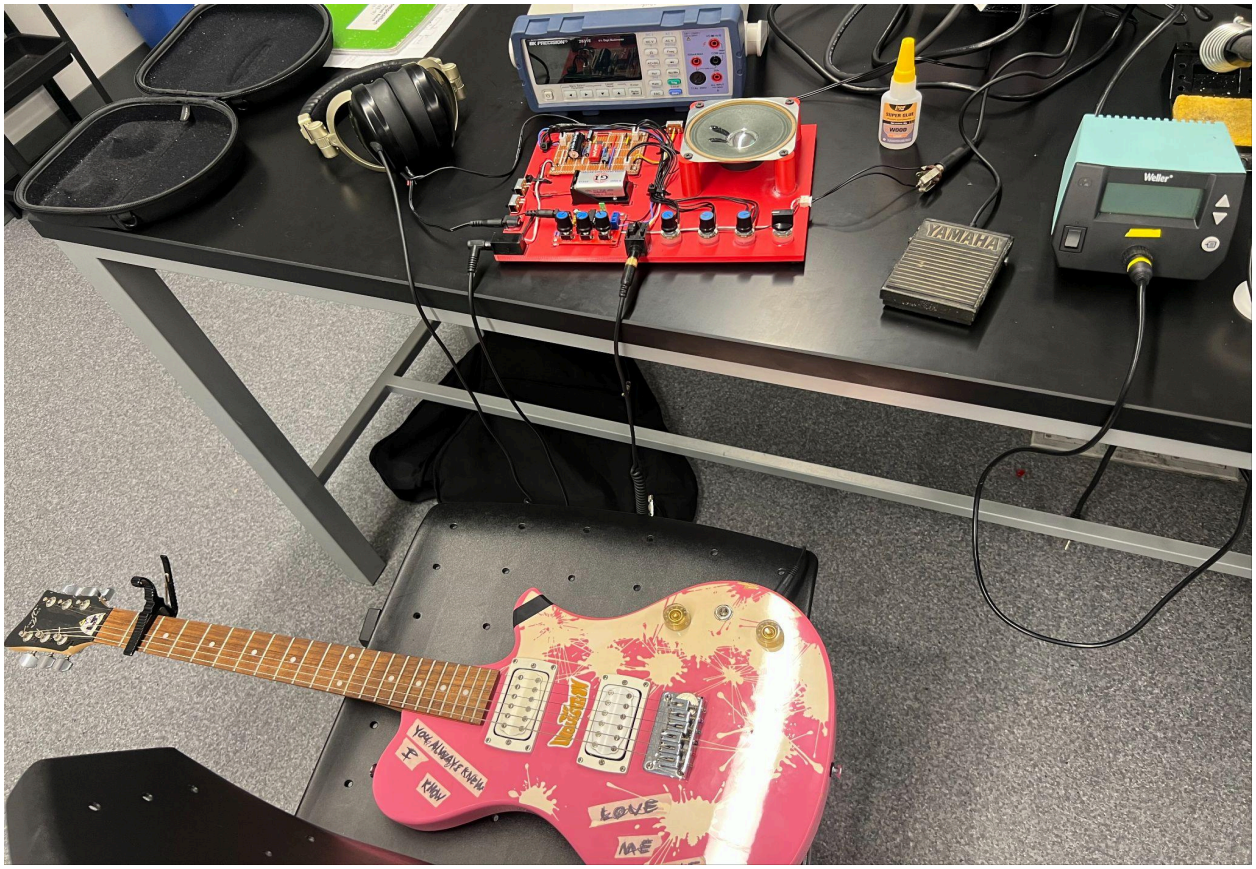
Data:

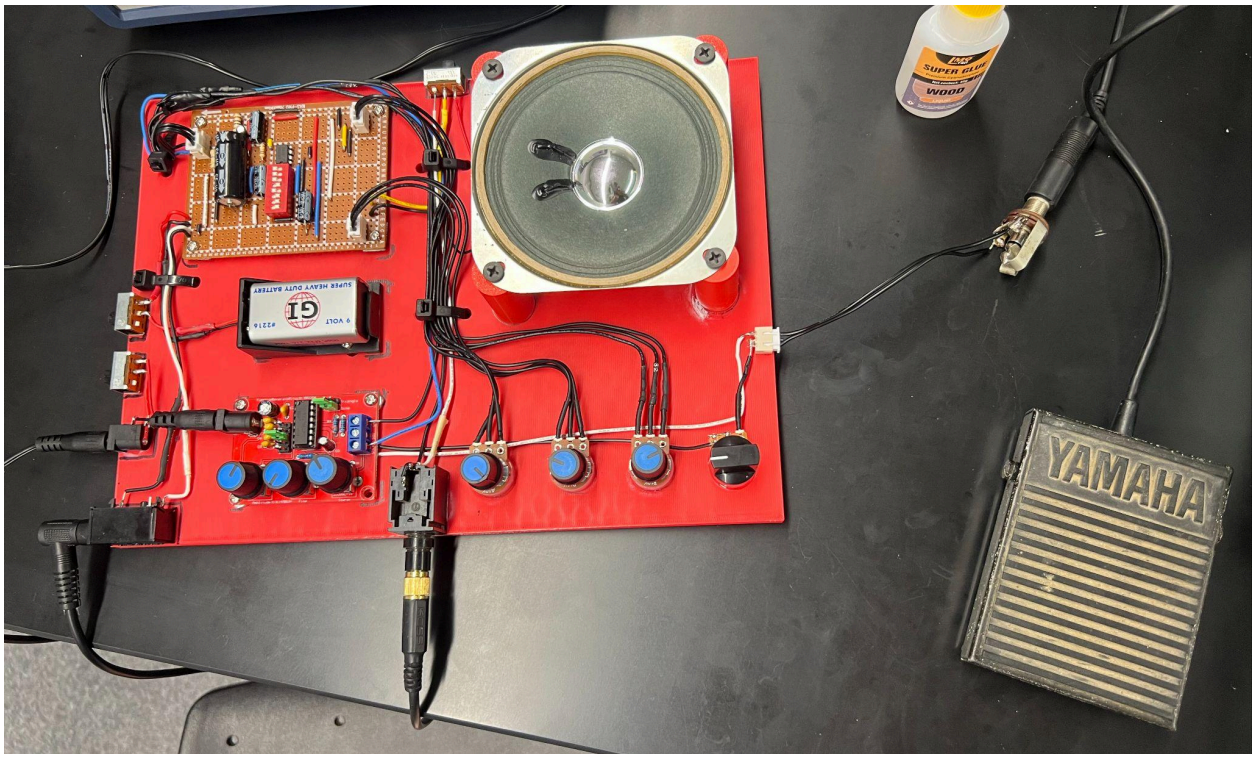
Finished Project Assembled:



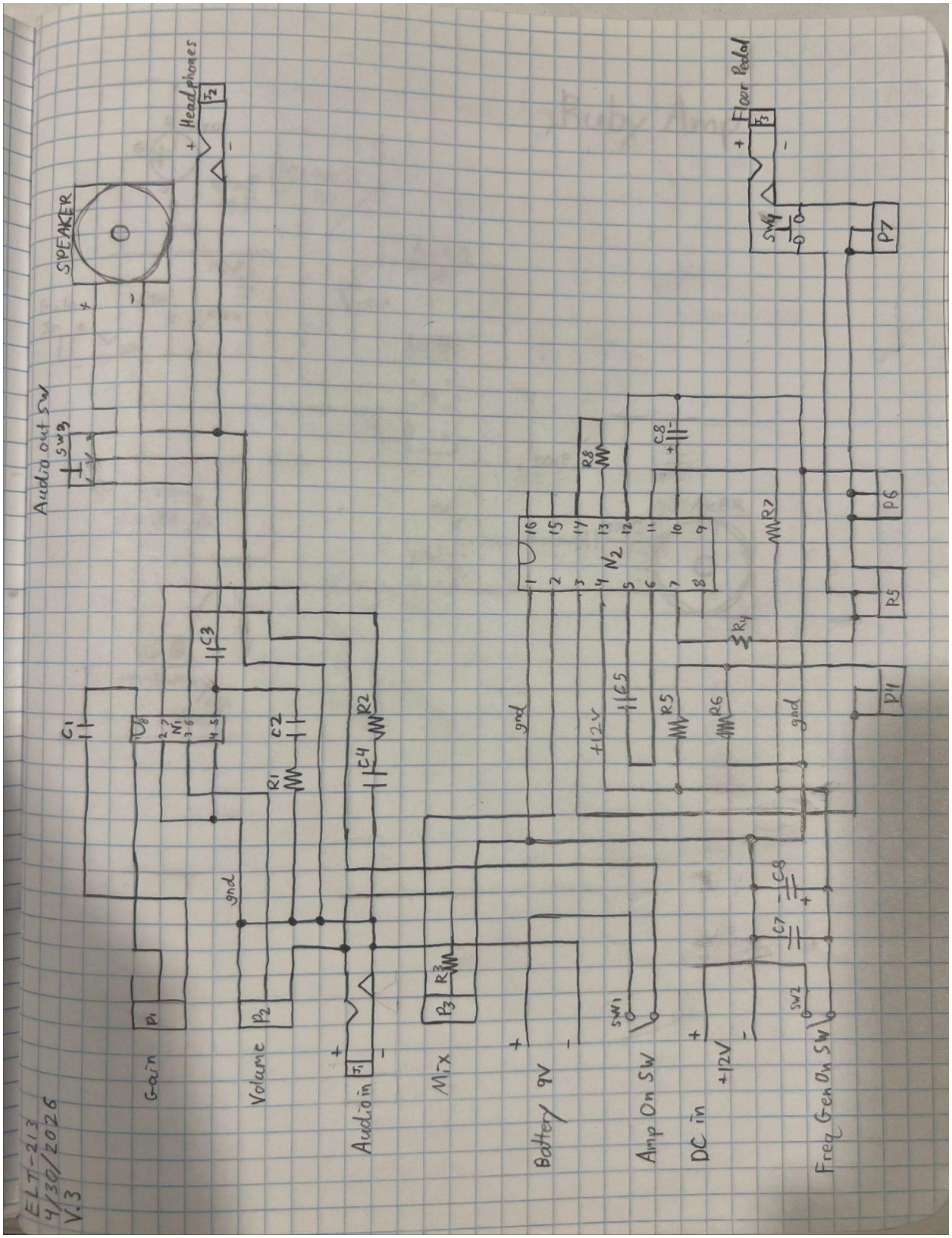


Finished Project With Floor Pedal Mod:

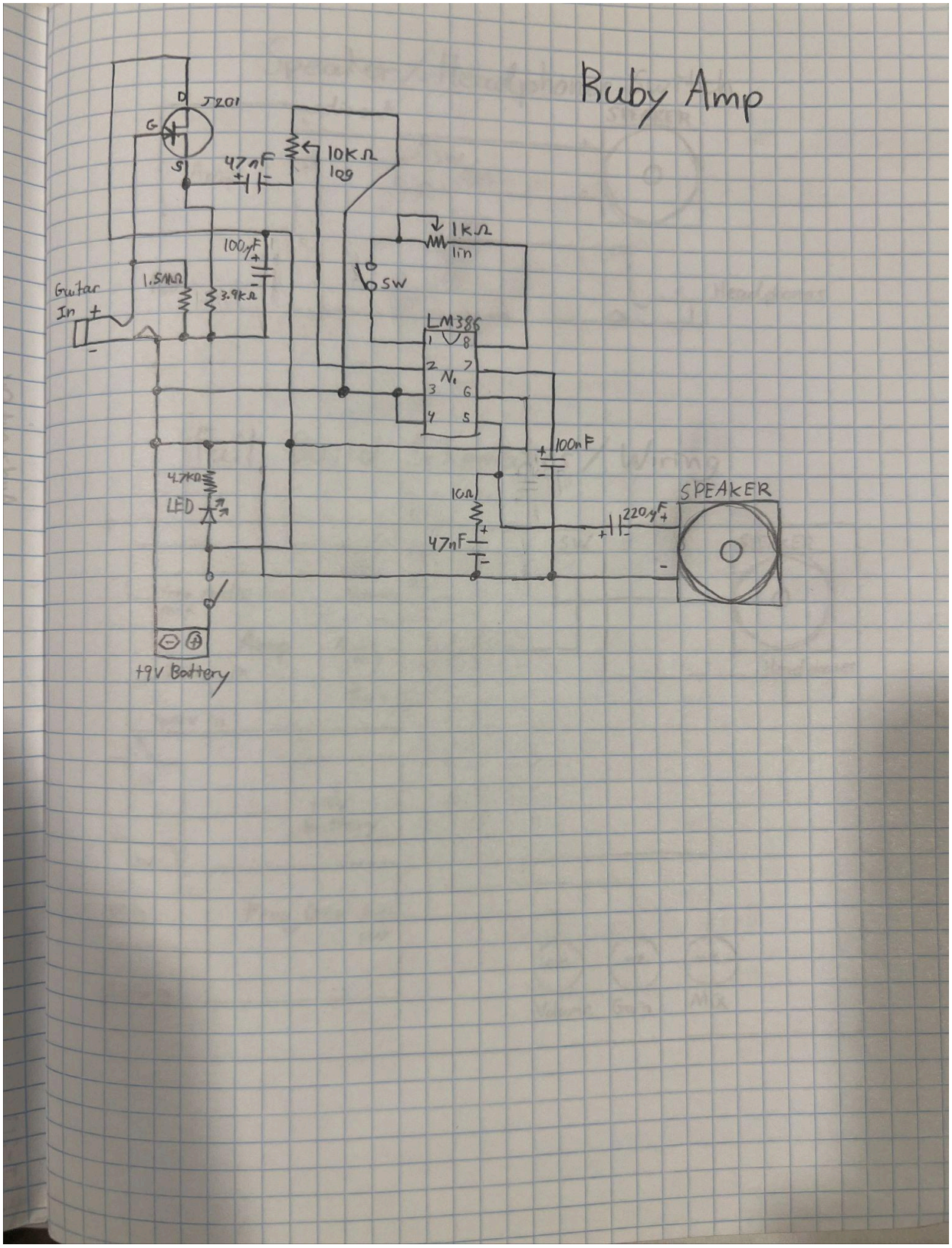




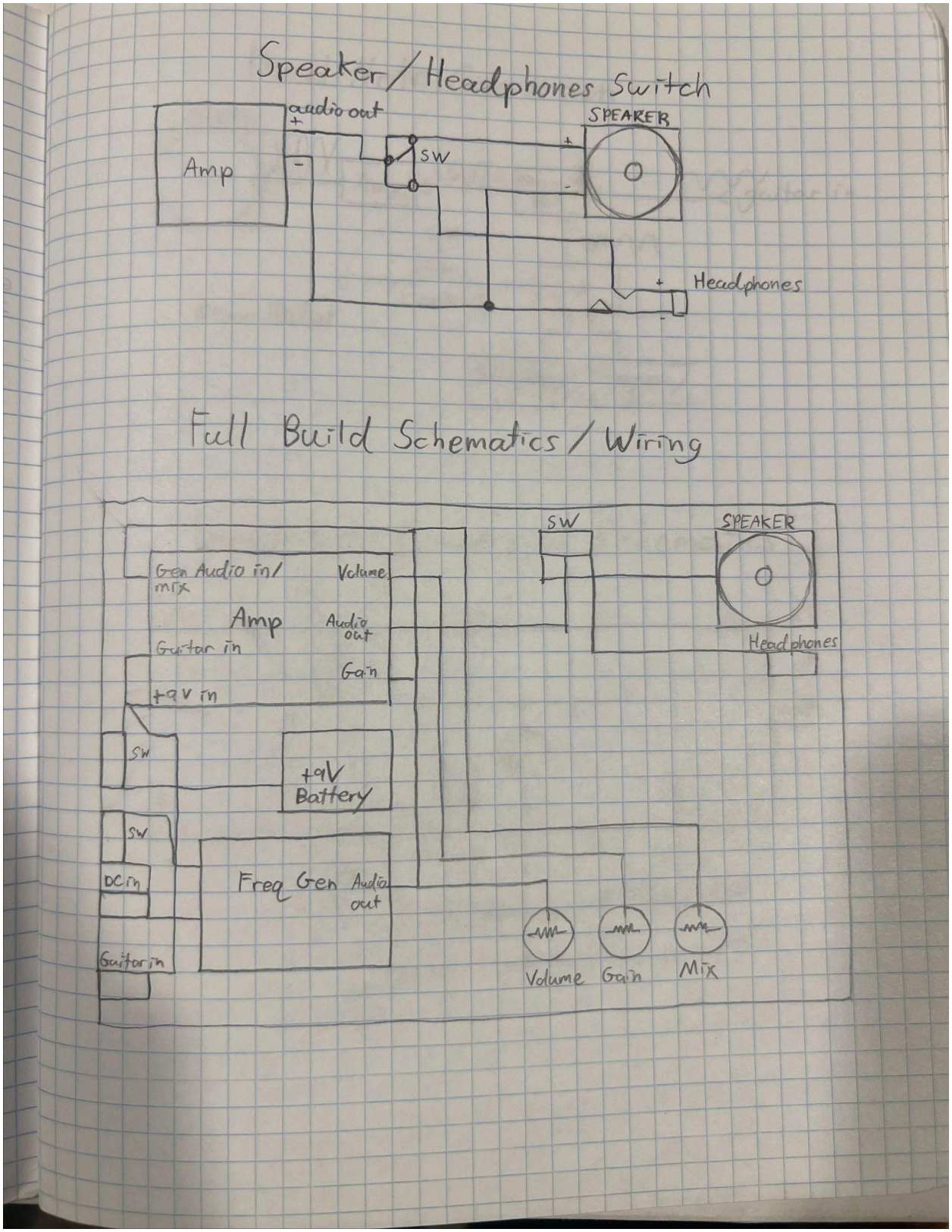
Final Schematics:



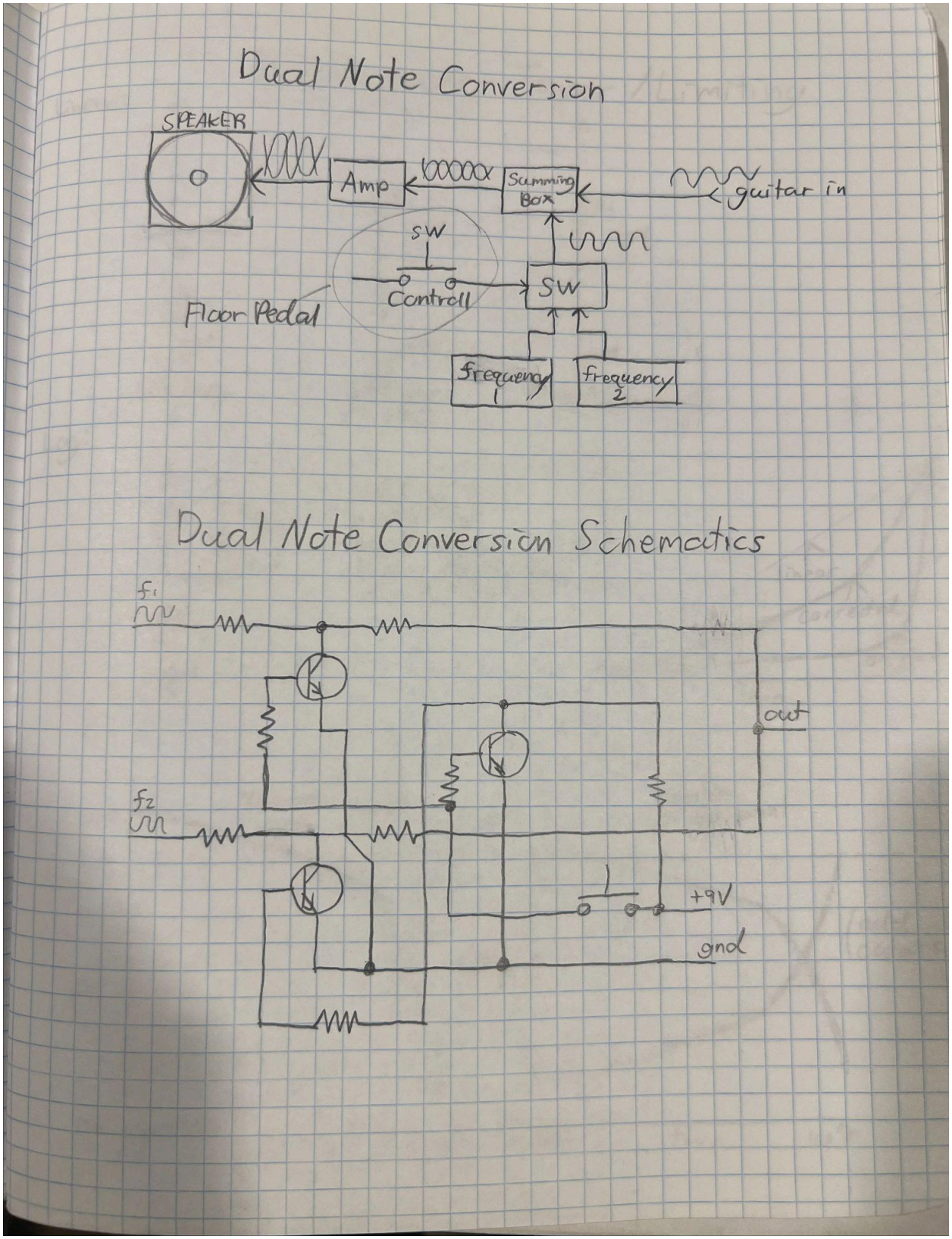
Ruby Amp Schematics:



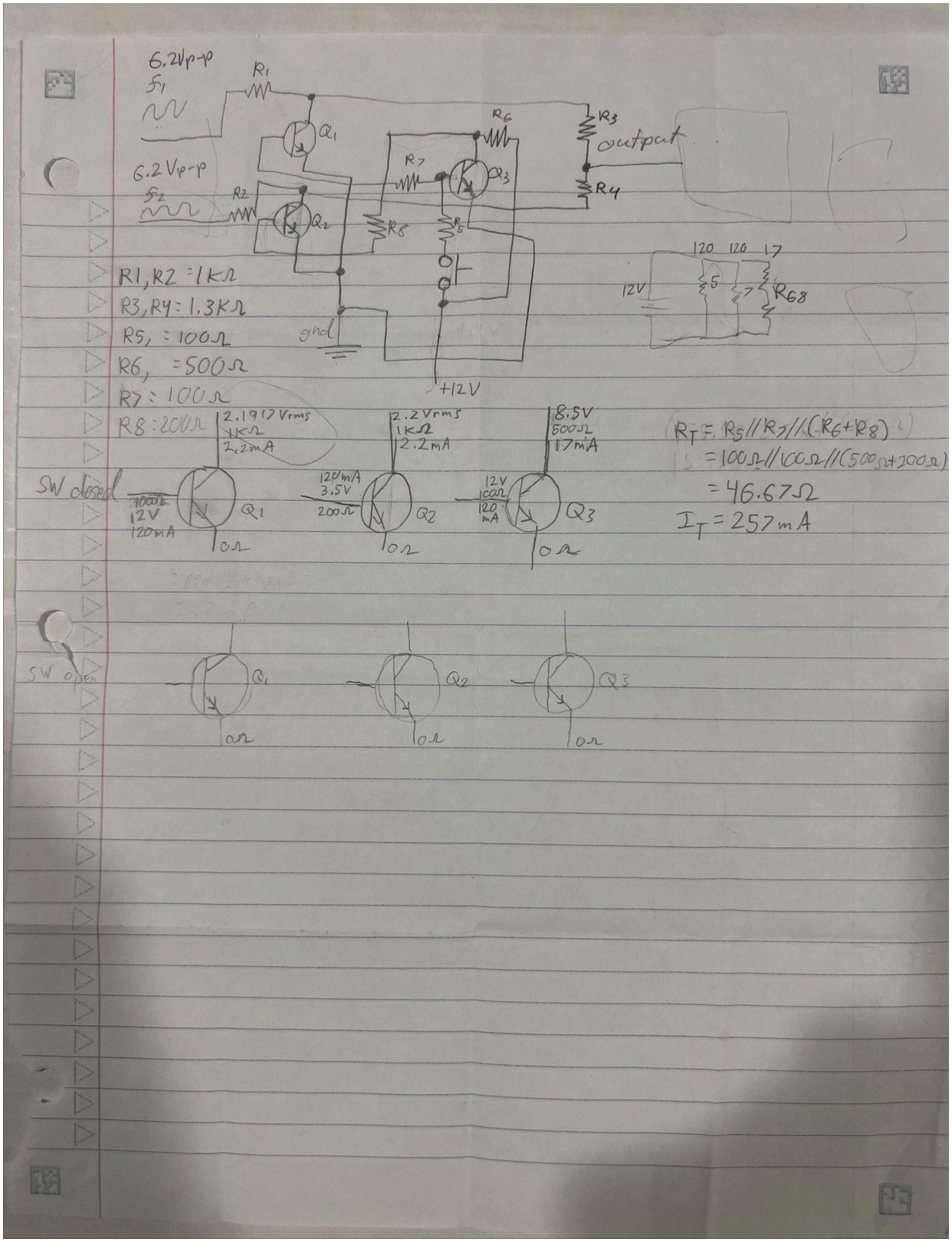
Wiring Diagram:



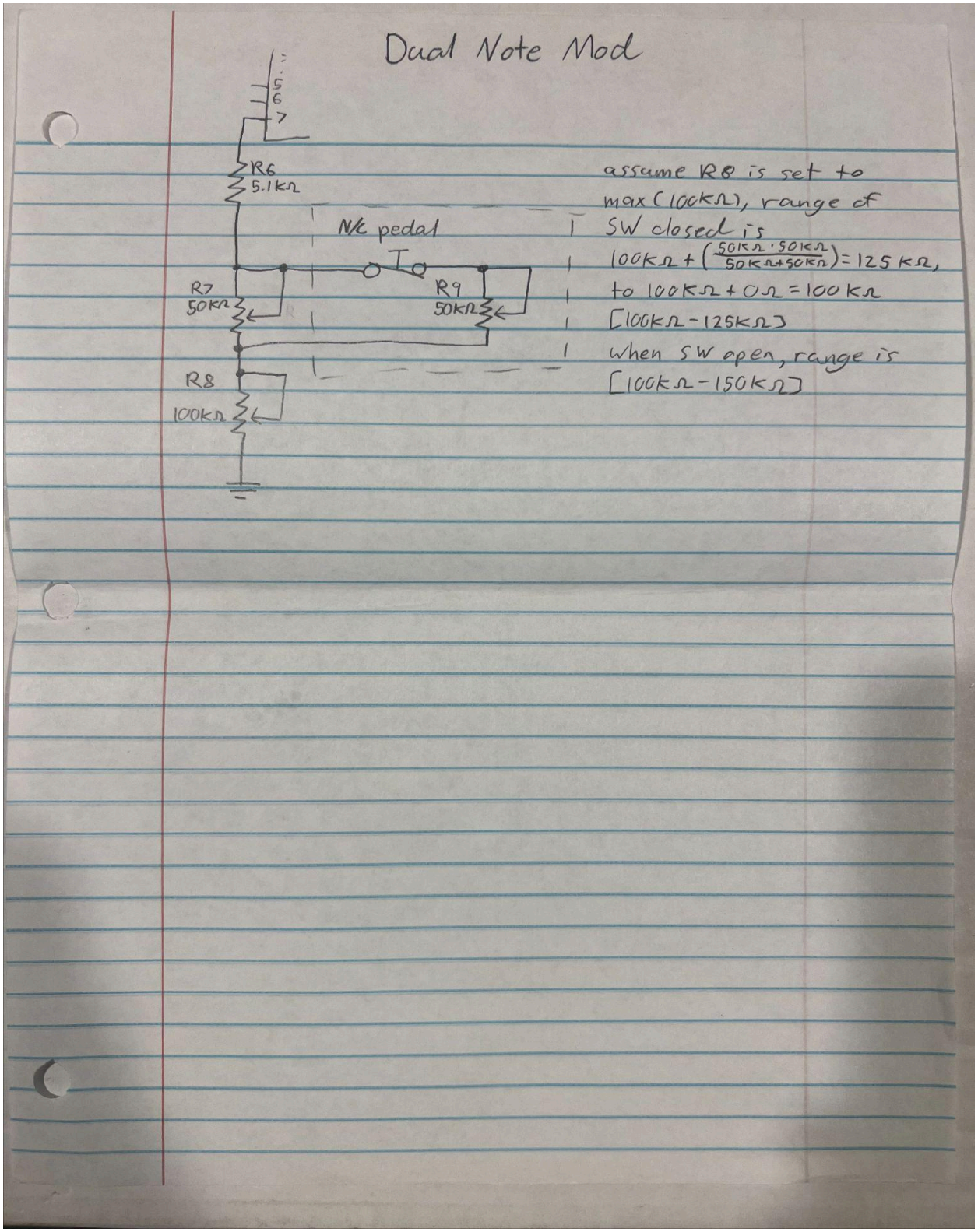
Dual Note Mod (1st Design):



Dual Note Mod Calculations (1st Design)



Dual Note Mod (2nd Design):



Final Conclusion:

This project achieved the original goal of building a self-playing guitar amplifier that produces a tunable drone note as well as normal guitar amplification. The design used a LM386 amp with a ICL8038 oscillator. However, the ICL8038 IC was delayed in shipping. An XR2206 kit was substituted and provided a stable 100 to 3,000 Hz range for the drone note.

During assembly and testing, each module was built on a breadboard and then transferred to a solderboard. Each module was also verified with a multimeter and oscilloscope. Several issues developed. These issues included: distortion from the over powered LM386 amp circuit, the oscillator output measuring near 8 volts peak-to-peak, signal cancellation when both sources were tied together, and ground loop humming from shared grounds. These were corrected by running the LM386 at 9 volts, removing the second LM386 amplifier, and adding a passive summing box with potentiometers and resistors to isolate and level-match the guitar and frequency generator inputs.

For final assembly, an open-faced 3D printed plate was used for easy access and wiring. Modifications were added to the design. These modifications included a floor pedal wired in parallel with the “fine” frequency control to switch between two preset drone notes. Along with the dual note floor pedal, a 3-pin toggle switch was installed to select a speaker or a headphone output.

Final testing confirmed the system worked as intended. The drone note was clear and adjustable, the guitar amplified cleanly, and the floor pedal switched notes reliably. This project was a huge success as it not only met all of the original design requirements, but it also exceeded them, giving musicians a new way to play and create their music.